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LIST OF ABBREVIATIONS

Abbreviation	Meaning
YOLO	You Only Look Once
CCTV	Closed-circuit television
PoE	Power Over Ethernet
PC	Personal Computer
LAN	Local Area Network
UN-SDG	United Nations - Sustainable Development Goals

Chapter 1

INTRODUCTION

Background of the Study

Motorcycle riders' safety has become the primary concern considering the increase in road accidents, emphasizing how important helmets are in preventing serious injuries and fatalities. Motorcycle accidents are currently the ninth most common cause of death in the Philippines, according to the Land Transportation Office. In addition, 98.7% of motorcycle-related traffic fatalities included drivers who were not wearing helmets, while 1.3% involved drivers who were wearing helmets (Sy, 2017). According to the World Health Organization (2023), the primary cause of death for motorcyclists is head trauma, and wearing a safe, high-quality helmet can lower the chance of brain damage by up to 74% and death by almost six times. Even with strict laws, compliance with motorcycle riders is still difficult, which calls for the development of comprehensive monitoring systems.

Given the high frequency of motorcycle accidents, ensuring that riders follow helmet requirements is crucial. Traditional means of monitoring helmet usage, including inspections by hand and stationary cameras, have limited coverage and efficiency. The use of technology to automatically recognize helmet usage on highways and roads is a possible tactic. Object recognition algorithms are often applied to evaluate preceding traffic data to identify high-risk regions for accidents or congestion. Object recognition algorithms, particularly YOLO, have shown outstanding potential in properly identifying various objects in real time, including helmets.

A significant factor in the high number of road accidents and injuries in the Philippines is motorcycle riders' persistent disregard for protocols requiring them to wear helmets. To ensure compliance and improve safety measures, a major difficulty is reliably recognizing and monitoring riders' helmet usage. Furthermore, it is important to identify which object recognition

algorithm effectively recognizes helmets. The key problem is the lack of an efficient and reliable method for monitoring and assessing helmet use across different locations, especially in high-risk areas.

With millions of unregistered motorcycles in the Philippines and over 8.5 million already registered, there are worries about their safety. Around 2016, Dr. John Juliard Go of the World Health Organization (WHO) Philippines reported that 11,264 people died in accidents on Philippine roads, with 5,970 (53%) of those killed being riders or passengers in motorcycles, tricycles, and other two- or three-wheeled vehicles; more than 90% in these people were not wearing on a helmet (Go, 2016, as cited in Tan, 2018). In addition, Sy (2017) reported that Central Visayas had the highest percentage of injuries reported (17.4%), followed by Western Visayas (10.6%) and Central Luzon (9.4%). These figures highlight the urgent need for concrete attempts to increase motorcycle riders' knowledge of traffic safety and encourage them to wear helmets. Schemes like thorough monitoring of helmet use nationwide have become critical because, without sufficient enforcement and monitoring methods, the danger of injuries and fatalities on Philippine roads will only increase.

According to Nantulya and Reich (2003), motorcycle riders' noncompliance with helmet usage can be related to a number of causes, such as lack of knowledge, poor enforcement, and riders' experiences with inconvenience. Socioeconomic considerations are particularly important because some riders may not be able to purchase expensive helmets or may choose to prioritize other costs. Dela Cruz et al. (2022) established that individuals who have enrolled in driving schools are more likely to wear helmets regularly compared to their counterparts, which implies that professional training and education can affect safety measures. It is very crucial to tackle these issues with care strategically because the danger of injuries and deaths on Philippine roads will just grow in the lack of productive safety measures.

The Helmet Recognition System with Algorithm Analysis suggests the creation of a system that implements high-level object recognition algorithms, e.g., YOLO, to solve these serious problems. The study attempts to identify the most effective approach for real-time helmet

recognition by carefully analyzing and comparing the performance of different algorithms. This device is strategically placed on roadways to continuously recognize and record motorcycle riders' helmet usage. Furthermore, to effectively locate high-risk areas, increase enforcement operations, and possibly decrease the rate of motorcycle-related injuries and fatalities, cooperation with local authorities will be necessary. The researchers hope to contribute to road safety throughout the Philippines and protect the lives of motorcyclists through the implementation of these preventive steps.

Objectives of the Study

The study's general objective is to aid local authorities and motorcycle riders in enhancing helmet compliance. The study aims to improve the effectiveness of safety monitoring systems. This research benefits barangay officials, local government units (LGUs), and law enforcement agencies by supporting efforts to promote safety awareness to reduce accident risks on roads.

Specifically, it aims to:

1. Design a recognition system that can accurately recognize motorcycle riders with or without helmets.

2. Develop an image and video recognition algorithm using an object recognition program.

    a. Implement data analytics tools that process the captured data and quantify the number of both with and without helmet motorcycle riders with the use of statistical methods and machine learning for results.

    b. Gather and analyze community data and assess adherence to road rules, as well as provide reports and insights to local computer systems, local authorities, and policymakers to improve road safety measures for motorists.

3. Test algorithm accuracy, precision, and recall of the system in the testing environment.

4. Evaluate the system's performance using ISO 25010:2011. Underlying this, these are the following: (1) functional suitability, (2) performance efficiency, (3) compatibility, (4) usability, (5) reliability, and (6) portability of the output system.

Scope and Limitations of the Study

Motorcycles are among the most popular forms of transportation in the Philippines, but a significant concern lies in the lack of helmet compliance among some motorcycle riders. This defies safety regulations and increases the risk of severe injuries in the case of accidents. To address this issue, the Helmet Recognition System (HRS) was developed as a potential technological solution.

This study aimed to evaluate the effectiveness of the Helmet Recognition System (HRS) in enforcing helmet use regulations for motorcycle riders in a controlled setting within the Philippines, and a specific HRS solution and its associated algorithm will be deployed to recognize helmet use. The study used the implementation of Algorithm Analysis implementation through CCTV cameras, whereas the area of the said community detects and analyzes if the condition is met. Recorded video footage was used to review the exact same purposes. The researchers developed a program that could detect motorists who have and have not used a helmet compared with traditional methods relying on human observation. Analyzing the program's efficiency and accuracy, it aimed to identify the most effective technological solution for enforcing helmet regulations and promoting motorcycle safety.

Furthermore, the effectiveness of any HRS hinged on the chosen algorithm and the quality of its training data, even in the controlled setting, might not account for factors like passengers on the motorcycle or poor weather conditions, which can all reduce the system's accuracy compared to the ideal scenario. It was important for the researchers to note that certain conditions could limit the system's accuracy. These included varying lighting conditions (e.g., nighttime, low-light environments, or glare) that hindered accurate detection. While the system was primarily designed for motorcycle detection, it may also detect bicycles, as they are similarly classified as two-wheeled vehicles. However, bicycles were not listed as motorcycles in the output, as the system differentiated between the two. Improving the detection accuracy for bicycles would require additional training of the algorithm using more diverse datasets.

Numerous advantages can be gathered from this study to different interest groups; the major recipients are motorcycle riders, who usually witness a considerable rise in helmet conformity because of accurate recognition and potential enforcement actions. This would mean fewer head injuries during accidents, making them safer for many individuals and families. Moreover, pedestrians on the roads can also benefit since there is a possibility of lower motorcycle crashes, resulting in safer public transportation. The use of more helmets by motorcyclists can minimize the risk of severe injuries for both riders and pedestrian victims in accidents, leading to safer general conditions for everybody using the road. Lastly, future researchers will benefit from this study as the information gathered and analyzed will be part of the growing body of literature on how HRS technology can enhance helmet use. Consequently, this knowledge base was used to devise more efficient and effective strategies aimed at enhancing motorcycle safety in the Philippines and other countries with the same set of concerns.

This study is aligned with the TUP R&D Agenda and UN-SDG number 9, “Infrastructure, Innovation, and Industry,” and 11, “Sustainable Cities and Communities.”

Significance of the Study

The role of automobiles, such as motorcycles, in daily life is undeniable, as they are commonly used for commuting to places such as work and school, as well as for leisure activities like dining out. However, it is a concerning reality that some individuals fail to comply with road rules and regulations, particularly in the Philippines, where helmets are often neglected, regardless of the distance traveled. This non-compliance increases the chances of events in case of an accident.

Checking and enforcing traffic laws is essential to reduce the number of instances of riders not wearing helmets while traveling. This study sought to address the need for tighter enforcement of regulations in terms of ensuring that individuals observe helmet-wearing mandates. While some areas have high compliance with the rules, most have a high percentage of violations of the rules. The researchers have analyzed data concerning the percentage of helmet

use in different locations with the help of object-oriented analysis software. In case the findings indicate a low level of compliance, then an enforcer will be placed there to continue monitoring the level of compliance and implement measures to reduce the failure percentage of individuals in observing their safety.

Chapter 2

CONCEPTUAL FRAMEWORK

This chapter reviews related literature and studies, providing the foundation for the research framework, programming software, components, and materials for system development. It also presents the conceptual model and defines key terms.

Review of Related Literature and Studies

This section presents research and studies supporting the project's conceptual framework. Figure 1, a Fishbone Diagram, outlines four key factors influencing the issue: Materials, Measurement, Mother Nature, and Machines.

Research Problem

This section delves into relevant research and studies that form the foundational basis for the project's conceptual framework. These studies help us comprehend the fundamental elements and dynamics influencing the issue at hand, which supports the project's methodology and strategy. The Cause-Effect Model outlines four primary factors—Materials, Measurement, Mother Nature, and Machine—that contribute to a broader social issue.

As indicated in Figure 1. These are essential to the performance and efficiency of the system. To deliver a complete response, they act in concert to create a web of interdependence. One must understand these relationships. To formulate targeted strategies to minimize their impact and optimize system performance as a whole.

Figure 1. The Cause-Effect Model

Figure 1 shows the main problem, which is that Motorcycle Riders do not follow the law by wearing safety gear and equipment, which is the helmet and still drive without one. The root cause can be clustered into four factors: materials, mother nature, measurement, and machine (Giron et al., 2020).

For materials, according to Aquitan et al. (2023), traditional methods of video surveillance rely on human observation, which is susceptible to fatigue and limited cognitive abilities when monitoring, which is one reason for alleviated surveillance and enforcement. Giron et al. (2020) also mention the lack of workforce in apprehending traffic violators, which riders of motorcycles can take a trip without wearing one.

Regarding the measurement, Dasgupta et al. (2019) emphasize the high speed of vehicles because these motorcycles do not appear clearly in the frames, making detection a very challenging task. Complexity, such as the motorcycle's shape, color, and size, is critical for the system to work properly. At the same time, Susa et al. (2022) mention multiple checkpoints. Checkpoints are one practice in the Philippines to ensure safety and regulate compliance with wearing helmets for motorcycle riders, but upon observation, it can cause traffic congestion on many public roadways.

Related cases about mother nature are only discussed by Talaulikar et al. (2019). They said that it is highly recommended that the placement of CCTV is surveyed properly to ensure optimal performance of the video surveillance and image.

Regarding the machine, Low-Quality Images and Video sources, according to Swapna et al. (2019), are necessary to use better equipment or sources to get the optimal result in recognizing whether motorcycle riders wear a helmet or not. Talaulikar et al. (2019) mention that the use of an offline system requires a large workforce and storage for video footage. It is hard for the system to maintain if this problem occurs, and minimizing the cost without losing performance as much as possible is appreciated.

The Importance of Helmet Use

It is impossible to underestimate the importance of helmet use in promoting road safety. According to various research, helmets reduce the chances of experiencing traumatic brain injuries can save lives.

Moreover, 91 of 180 countries have implemented national laws against the major risk factors, including speeding, driving under the influence, seat belts, and child restraints, but not wearing motorcycle helmets (Bachani et al., 2017). Still, not everyone wears them, which emphasizes the need for constant reminders and the enforcement of helmet laws. Helmets are very important in protecting motorcyclists from severe head injuries and fatalities in the event of an accident.

The Role of Helmets in Reducing Fatalities and Injuries

Head injuries are the leading cause of fatality among motorcycle riders. They are involved in 70% of motorcycle accidents (Abbas et al., 2012). Wearing a helmet is essential to averting such catastrophes since it protects the brain and absorbs force. According to study, motorcycle riders who do not wear helmets have a 40% higher risk of dying from a head injury than those who do.

A 10% increase in helmet use can save one life per 1,000,000 people per year, according to Abbas et al. (2012). It has been demonstrated that pursuing helmet infractions reduces the likelihood of traumatic brain damage. Wearing safety-standard helmets is also crucial to help reduce injuries and fatalities. More people riding motorcycles and speeding may be the cause of higher fatality rates in countries with helmet laws.

Legal Frameworks and Policies on Helmet Use in the Philippines

All drivers and motorcycle riders must use approved helmets on all Philippine highways and roads by virtue of the Motorcycle Helmet Act of 2009 (Republic Act No. 10054). By decreasing the number of motorcycle accidents involving head injuries, which are the leading cause of death in these accidents, the act hopes to enhance road safety. Safety helmet standards are provided by the Department of Trade and Industry (DTI). For a helmet to be deemed

compatible, it should bear either the Philippine Standard (PS) mark or the Import Commodity Clearance (ICC) mark.

Penalties for non-compliance with the helmet requirement are also stipulated in Republic Act No. 10054, with the second-time offenders paying higher fines. Retailers who supply or sell substandard helmets are also subject to sanctions. The Act places importance on the necessity of raising public awareness, which necessitates an extensive information campaign to inform the public about the advantages of wearing a helmet in terms of saving lives. To minimize motorcycle fatalities and enhance public road safety, the above regulations should be enforced strictly.

Challenges in Helmet Use Enforcement

Helmet use enforcement is greatly hindered by major challenges, especially in the form of myths that exist, misuse of helmets, and the lack of a proper monitoring and detection system. Most motorcycle riders do not wear helmets correctly, as per UN ESCAP, 2020. Most individuals wear their helmets improperly or do not buckle them up. Ignorance and poor community programs are the reasons, especially in areas with stringent funds. Moreover, myths and misconceptions relating to helmets are a prominent factor in the denial of their usage by people. In fact, some riders are against the use of helmets owing to discomfort or cultural needs such as religious head cover or hair styles, and others believe that they hinder ear function or neck injury. The ineffectiveness of surveillance mechanisms and the insufficient attestation to child passengers is another major hindrance to the enforcement of helmet mandates. The UN ESCAP (2020) indicates that a low rate of helmet adoption among children is due to several ESCAP member countries having few or no laws regarding child passengers.

There are a few challenges when utilizing CCTV cameras to determine whether individuals are wearing helmets. Some of these are that it can be a bad lighting situation, it can be a bad weather situation, and it is not easy to distinguish clearly whether an individual is wearing a helmet on a moving car from various directions. Due to these difficulties, we require stringent

enforcement and effective public education campaigns to ensure individuals wear helmets.

Manual Monitoring and Enforcement of Helmet Use

Studies on road safety always point out the necessity of tougher enforcement but are sometimes devoid of detailed proposals for enhancing road observation methods (Shuey, 2019).

Enforcement and monitoring helmet wear are key to motorcycle safety, and periodic inspections are needed to guarantee that riders properly wear helmets and habitually do so. Police officers may carry out such inspections via roadside stops or watchtower cameras in high-risk zones. However, depending on human inspection to a large degree has its disadvantages, including limited resources and personnel, which lead to sporadic and less effective enforcement.

Therefore, integration of new technology and innovative methodologies is required to strengthen human inspection mechanisms and increase enforcement effectiveness. This could entail automatic helmet detection software in neighborhoods or via surveillance efforts.

Real-Time Automatic Detection of Motorcyclists with and without a Safety Helmet

Researchers at Padmini Velagapudi Ramakrishna Siddhartha Engineering College et al. (2020) established a new system which employs real-time monitoring in order to automatically detect helmet-less motorcyclists. This gadget makes the roads safer by detecting strengthening enforcement and minimizing the requirement for hand monitoring, resulting in improved helmet law compliance. The researchers extended on current CCTV infrastructure and created a modern helmet detection system using machine learning and image processing technologies.

Automatic Detection of Motorcyclists without Helmet

India, the second-most populous nation in the world and tenth-largest economy, depends overwhelmingly rely on motorcycles as a cheap and accessible form of transport. Still, the great possibility of severe injury in motorcycle collisions highlights the necessity of helmet use. Despite the presence of helmet legislation, enforcement issues have resulted in demands for better compliance.

In 2013, the Surat Municipal Corporation pioneered the use of real-time CCTV monitoring integrated with a central command and control center to enforce helmet use. This system is complemented by the E-Challan system, which electronically issues fines to violators, leveraging technology to enhance road safety and improve adherence to traffic rules.

Advanced Image Processing Techniques

Digital image processing, unlike traditional methods, uses algorithms to effectively manipulate images. This is required in the detection of helmets with a vast amount of visual information. Haar Cascades applies a series of classifiers to detect particular features of an image, for instance, helmets. Detection speed and accuracy are enhanced with this method by focusing on only the significant image features. By prioritizing primary attributes, Haar Cascades can work on images effectively even for real-time processing, without expending too much computational effort. The method is applicable in object detection operates where time and precision are critical, such as in surveillance systems.

Object Detection Based on Deep Learning

Deng et al. (2020) created a new system to enhance road safety by resolving the problem of helmet-less motorcyclists. Their system employs real-time video surveillance to automatically identify helmet-less riders and enforce traffic law more effective. This study was prompted by the significant role helmets have in avoiding head injuries and deaths in motorcycle crashes. However, helmet laws may be difficult to enforce and require a lot of resources.

The system can be easily interfaced with existing CCTV systems commonly used in urban locations. The system can identify helmetless riders quite effectively in real-time by live video analysis. This not only improves the efficacy of law enforcement but also reduces the need for manual surveillance, allowing authorities to make better use of their resources.

Two-stage Target Detection Framework

R-CNN

In 2014, R-CNN was introduced as one of the pioneering object detection models utilizing convolutional neural networks as shown in figure 2. The improved version of R-CNN achieved a mean Average Precision (mAP) of 66%. This model employs Selective Search to generate approximately 2000 potential object regions per image for detection. These regions are uniformly resized to create fixed-length feature vectors, which are then classified by a Support Vector Machine (SVM). The SVM assigns class labels to the detected regions based on the learned features. Finally, a linear regression model is trained to refine the bounding boxes of the detected objects.

[FIGURE 2: R-CNN architecture pipeline — 1. Input image -> 2. Extract region proposal (~2K) -> 3. Compute CNN features -> 4. Classify regions (example outputs: cat? no, dog? yes, pig? no)]

Figure 2. R-CNN architecture

Source: https://iopscience.iop.org/article/10.1088/1742-6596/1684/1/012028/pdf

SPP-Net

In 2015, the Spatial Pyramid Pooling (SPP) model was introduced to address the limitations of R-CNN as shown in figure 3, namely low detection efficiency and the requirement for fixed-size input regions. SPP improves feature extraction by processing region proposals directly on the feature map after a single pass through the convolutional layers, thereby reducing redundant computations. While SPP-Net outperforms R-CNN by performing feature extraction on the entire image just once, thus avoiding repetitive computations, it inherits similar drawbacks.

1. Complex multi-step training procedures and;

2. The need for training separate SVM classifiers and additional regressors.

Despite these improvements, the complexity of SPP-Net limits its usability for end-to-end learning frameworks. The model requires careful coordination between feature extraction, classifier training, and bounding box regression, making the implementation process time-consuming. Additionally, the reliance on separate components introduces potential inefficiencies and complicates the integration of updates or enhancements to the model.

[FIGURE 3: SPP-Net architecture — input image -> convolutional layers -> feature maps of conv (arbitrary size) -> spatial pyramid pooling layer -> [16x256-d, 4x256-d, 256-d feature representations] -> fixed-length representation -> fully-connected layers (fc, fc)]

Figure 3. SPP-Net architecture

Source: https://iopscience.iop.org/article/10.1088/1742-6596/1684/1/012028/pdf

Fast R-CNN

The Fast R-CNN introduces three significant modifications compared to its predecessor as shown in figure 4, R-CNN. Firstly, it replaces the SVM classifier used in R-CNN with a softmax function for classification. Secondly, inspired by SPP-Net, the model incorporates a region of interest pooling layer to replace the last pooling layer in the convolutional network, enabling the transformation of candidate box features into a fixed-size feature map for access to the fully connected layers. Lastly, the last softmax classification layer of the CNN network is substituted with two parallel fully connected layers. Despite these advancements, Fast R-CNN still falls short of meeting real-time detection requirements. The high computational cost of processing numerous region proposals and reliance on external generators like Selective Search limit Fast R-CNN's suitability for real-time applications, as it struggles to balance accuracy with speed.

[FIGURE 4: Fast R-CNN architecture — input image -> Deep ConvNet -> Conv feature map with RoI projection -> RoI pooling layer -> FCs -> RoI feature vector -> outputs: softmax (classification) and bbox regressor (for each RoI)]

Figure 4. Fast R-CNN architecture

Source: https://iopscience.iop.org/article/10.1088/1742-6596/1684/1/012028/pdf

Faster R-CNN

A new model was introduced in 2015, the Faster R-CNN model as shown in figure 5. This model improves upon its predecessors by using region proposal networks instead of the Selective Search method for generating region proposals. It comprises two modules: one fully convolutional neural network generates all region proposals, while the other implements the Fast R-CNN detection algorithm. These modules share a set of convolutional layers. When an input image is processed through the CNN network, it produces two outputs: a feature map for the RPN network and a higher-dimensional feature map at a specific convolutional layer.

This architecture allows for end-to-end training, making it faster and more efficient compared to previous models. As a result, Faster R-CNN significantly reduces the time required for object detection tasks while maintaining high accuracy, and its integration of the region proposal network improves the quality and speed of detecting objects in images. As a result, Faster R-CNN significantly reduces the time required for object detection tasks while maintaining high accuracy.

[FIGURE 5: Faster R-CNN architecture — image -> conv layers -> feature maps -> Region Proposal Network (proposals) -> RoI pooling -> classification loss & bbox regression loss]

Figure 5. Faster R-CNN architecture

Source: https://iopscience.iop.org/article/10.1088/1742-6596/1684/1/012028/pdf

One-stage Target Detection Algorithm

YOLOv1

YOLOv1 does not need a separate region proposal extraction step, unlike other models. It simplifies the detection process with a straightforward CNN network structure. As shown in figure 6, its main concept involves feeding the entire image directly into the network and obtaining bounding box locations and categories directly at the output layer. The image is divided into an SS grid, where each grid cell predicts B bounding boxes and confidence scores for these boxes, resulting in a total prediction of B(4+1) values per cell. YOLOv1 achieves impressive real-time detection speeds, up to 45 frames per second on a single TitanX GPU. These speeds make it particularly suitable for time-sensitive applications such as autonomous driving, video surveillance, and robotics.

[FIGURE 6: YOLOv1 architecture — sequence of convolutional and maxpool layers operating on a 448x448x3 input image, progressively reducing spatial dimensions through 112, 56, 28, 14, 7 sizes and increasing channels (192, 256, 512, 1024) before flattening to 4096 and outputting 30 detection values per grid cell.]

Figure 6. YOLOv1 architecture

Source: https://iopscience.iop.org/article/10.1088/1742-6596/1684/1/012028/pdf

YOLOv2

YOLOv2 adopts a novel fully convolutional feature extraction network called Darknet-19, comprising 19 convolutional layers and five maximum pooling layers. By integrating batch normalization layers into the convolutional layer, removal of dropout, introducing an anchor box mechanism, utilizing k-means clustering on training set bounding boxes, and implementing multi-scale training, YOLOv2 achieves notable improvements in recall and accuracy.

YOLOv3

This model stands out as the most balanced object detection model, excelling in both detection speed and accuracy. Regarding category prediction, YOLOv3 transitions from single-label to multi-label classification, replacing the original softmax layer with a logistic regression layer. It uses different scales to predict, using a method like FPN (Feature Pyramid Network) to combine them. The network architecture adopts a deeper feature extraction network, Darknet-53. While YOLOv3 further enhances detection speed and significantly improves the detection of small targets, it does not exhibit substantial improvements in detection accuracy, particularly when IOU (Intersection over Union) exceeds 0.5.

SSD

Inspired by the YOLO algorithm's regression concept and the anchor box idea from the Faster R-CNN detection model, SSD aims to enhance multi-scale object detection effectiveness as shown in figure 7. It achieves this by utilizing both bottom and high-level feature maps for detection. The fundamental architecture is based on VGG, with convolutional layers replacing the last two fully connected layers. This allows SSD to make predictions at multiple scales, improving its ability to detect objects of varying sizes within an image.

[FIGURE 7: SSD architecture — two grid representations showing default boxes overlaid on grid (left) and localization (loc: Δ(x_c, y_c, x_w, y_h)) and confidence (conf: (C_1, C_2, ..., C_p)) outputs from grid cells (right)]

Figure 7. SSD architecture

Source: https://iopscience.iop.org/article/10.1088/1742-6596/1684/1/012028/pdf

YOLOv4

YOLOv4 incorporates several enhancements to the original YOLO detection framework. These include Weighted Residual Connection, Cross Stage Partial connection, Cross mini-Batch Normalization, Self-adversarial training, Mish activation, Mosaic data augmentation, DropBlock, and CIou. The backbone network chosen is CSP Darknet53, to which an SPP module is added to expand the receptive field and isolate crucial contextual features. YOLOv4 also replaces FPN with PANet for path aggregation while maintaining the head structure of YOLOv3.

YOLOv5

The fifth iteration of YOLO, known as YOLOv5, was proposed by Ultralytics in 2020. Its speed and accuracy exceed those of all prior iterations. The YOLOv5 algorithm adjusts the width and depth of the backbone network using the depth_multiple and width_multiple parameters to produce four different model versions: YOLOv5s, YOLOv5m, YOLOv5l, and YOLOv5x. The most basic version, YOLOv5s, has the shortest model parameters and the quickest detection speed. Figure 2 shows its network structure, which consists of the focus, CSP1_x, CSP2_x, Conv-Bn-Leakyrelu (CBL), and spatial pyramid pooling (SPP) modules.

Detection System of Motorcyclists for Using Helmet

Motorcycles are among the most used vehicles on the road, and they are known for their popularity in maneuvering through traffic due to their narrow build. But at the same time, it is one of the most accident-prone vehicles, with human error being a leading cause. Wearing a helmet is so essential, as it can save lives in the event of an accident. The researcher addresses the critical issue of road safety and passengers, prompting the development of these detection systems. Aquitan et al. (2023) demonstrated the use of YOLOv8 technology for real-time helmet detection in traffic surveillance systems. The researchers aimed to develop a robust and reliable model suitable for practical deployment to enhance safety. In the Philippines, the “No Helmet, No Ride” law (Republic Act No. 10054) mandates helmet use for motorcyclists, but enforcement challenges arise due to inadequate surveillance and monitoring. To address these challenges,

Giron et al. (2020) proposed utilizing artificial intelligence, specifically computer vision techniques such as image classification and object detection, through CCTV systems to automate traffic enforcement.

[FIGURE 8: Sample Images from the Dataset — four surveillance camera images showing motorcycles and vehicles in various traffic scenes (daytime and nighttime)]

Figure 8. Sample Images from the Dataset

Source: https://tinyurl.com/4hkjxktr

According to Giron (2020), the increasing use of CCTV systems has made apprehending riders who violate helmet regulations easier. The study used Convolutional Neural Networks (CNNs) to identify whether riders use a helmet or not and use a high number of images to train for the experiment and its example can be seen in Figure 8.

[FIGURE 9: Dataset annotation of an authorized and unauthorized helmet — close-up annotated image of a purple helmet, and a wider view of a person on a motorcycle with the helmet annotated]

Figure 9. Dataset annotation of an authorized and unauthorized helmet

Source: https://tinyurl.com/56smsw5c

Another researcher utilizes an alternative version of YOLO (You Only Look Once), specifically YOLOv3, in a video-based system developed to determine whether riders are wearing helmets authorized by the law for use on roads and national highways. This system, exemplified in Figure 9, addresses the challenges often faced by the Metropolitan Manila Development Authority (MMDA) in enforcing helmet regulations (Susa et al., 2022). The developed system supports and simplifies the enforcement process, making it more efficient for authorities and manageable.

A similar case occurs in other countries, such as India, where motorcycles are widely used. Researchers developed methods to automatically identify riders who are not wearing helmets. According to Talaulikar et al. (2019), as shown in Figure 10, their approach involves capturing video feeds from surveillance cameras deployed along the roads and applying background subtraction techniques to identify moving vehicles. This enables the detection of

helmets from motorcycle riders, which is further enhanced through different machine-learning techniques.

[FIGURE 10: Generic framework — Video Stream -> Key Frames Extraction -> Background Subtraction (Identifying the Vehicular Objects) -> Motorcycle Objects Identification -> Helmet Detection using PCA and Machine Learning Techniques]

Figure 10. The generic framework of the proposed approach

Source: https://tinyurl.com/ykhepn4v

Swapna et al. (2019) employed image-processing techniques to identify motorcyclists not wearing helmets. Upon detecting a helmet violation, the system also captures and records the motorcycle's license plate number.

[FIGURE 11: Images of riders with and without helmets — two motorcyclists with bounding box annotations highlighting the head/helmet area]

Figure 11. Images of riders with and without helmets

Source: https://tinyurl.com/296ujrzj

As displayed in figure 11, the photographs of helmeted and non-helmeted riders, as cited by Swapna et al. (2019). It reflects the sharp distinction between motorcyclists dressed helmets and non-helmets, which is necessary for identifying helmet compliance through image processing. This differentiation aids in the identification of riders not adhering to safety regulations.

[FIGURE 12: Captured plate number image — close-up of a red motorcycle's license plate "TS09EZ.0521" with green bounding box annotation]

Figure 12. Captured plate number image

Source: https://tinyurl.com/296ujrzj

Figure 12 presents an illustration of a number plate image taken by a surveillance camera. As shown by Swapna et al. (2019), the quality of the captured image is vital to the number plate recognition process, as it affects the accuracy of the detection algorithm directly. High-quality images facilitate better identification of the number plate, which increases the capability of the system to identify the vehicle correctly. This figure highlights the importance of high-resolution, clear images in guaranteeing good vehicle detection and tracking.

According to Dasgupta et al. (2019), using the YOLOv3 model is the state-of-the-art method to detect motorcycle riders. Following these initial detections, a Convolutional Neural Network (CNN) is then applied as an additional layer specifically for helmet detection, as illustrated in Figure 13.

[FIGURE 13: Different phases of the proposed work — Motorcycle and multiple riders detection using YOLOv3 -> Crop the riders top one-fourth portion -> Helmet detection using proposed CNN -> No helmet / Helmet output classification]

Figure 13. Different phases of the proposed work

Source: https://tinyurl.com/mrx922m9

As shown in Table 1, the researchers examined various empirical literature that are related to the current study. Specifically, the researchers identified the strengths, weaknesses, opportunities, and threats (SWOT), which have been used as the basis for the justification of the proposed project study.

Table 1

SWOT Matrix

Title, Author, Year	Problem	Methodology	Strength	Weakness	Opportunity	Threat
Motorcycle Rider Helmet Detection for Riding Safety and Compliance Using Convolutional Neural Networks (Giron et al., 2020)	Classifying riders between wearing helmets or not	Convolutional Neural Network-Based recognition system	Variety of usable detection algorithm	Differences of reliability and accuracy	Ability to determine the most reliable and accurate algorithm	Existence of similar algorithms that may be better
YOLOv8-Based Real-Time Helmet Detection for Enhanced Traffic Monitoring and Surveillance (Aquitan et al., 2023)	Critical issue of road safety particularly with motorcycle riders	YOLOv8-based recognition system	Handles larger and diverse datasets and its processing and analytical power	Inability to detect a fast-moving object	Customized datasets can be trained based on objective	Variety of YOLO versions and adaption to new one needs to retrain
An Efficient Safety and Authorized Helmet Detection Using Deep Learning Approach (Susa et al., 2022)	Difficult to enforce the law regarding wearing authorized helmet	YOLOv3-based recognition system	Speedy identification and accuracy level	Complex to train, research needs 25 layers of training	Greater number of trainings will improve the detection greatly	Upcoming and new existing YOLO models

Title, Author, Year	Problem	Methodology	Strength	Weakness	Opportunity	Threat
A Hybrid Approach for Helmet Detection for Riders Safety Using Image Processing, Machine Learning, Artificial Intelligence (Swapna et al., 2019)	High rate of traffic rule violators who are not wearing helmets	OpenCV-based recognition system	Capable of processing and handling large datasets	Accuracy might decline in poor lighting conditions	Can integrate with advanced safety features, like collision detection	Adapting to new versions or image processing algorithms may require additional retraining of the helmet detection models
An Enhanced Approach for Detecting Helmet on Motorcyclists Using Image Processing and Machine Learning Techniques (Talaulikar et al., 2019)	Helmet non-compliance which leads to serious injuries	Helmet Detection using PCA and Machine Learning Techniques	High accuracy and reduced processing load through principal component analysis (PCA) and key frame selection	Reliant on selecting only key frames which may lead to missed detections if critical information is present in frames that are not selected	Can be connected to automatic ticketing systems	Ongoing advancements in detection algorithms may require regular retraining and adaptation
Automated Helmet Detection for Multiple Motorcycle Riders Using CNN (Dasgupta et al., 2019)	High rate of traffic rule violations particularly non-compliance of wearing helmets	YOLOv3-based recognition system for motorcycle riders and Convolutional Neural Network-Based recognition system for helmet detection	Excels at detecting a variety of classes, specifically motorcycles and people	Implementation of a non-GPU setup which may limit real-time application	Can be extended to detect other violations	Reliance on non-GPU hardware may limit performance in detection

Recognition systems are one of the great discoveries that are greatly appreciated in the modern era. A variety of algorithms were created to solve manual monitoring and data gathering. It speeds up the tallying of specific to complex data of object recognition. Researcher Giron et al. (2020) compare three (3) existing CNN (Convolutional Neural Network) such as MobileNetv2, Inception_v2, and FasterRCNN.

Table 2 shows the different outputs of different CNN models, and researchers found that FasterRCNN has a lower average in terms of loss or inaccuracy.

Table 2

Training details in different models (Giron et al., 2020)

Models	Average sec/step	Time hour	Steps	Inference graph size	Loss

ssd_mobilenet_v2	1.5	6.75	16127	47.9MB	2.4
Ssd_inception_v2	1.7	7.5	16214	155MB	2.5
Faster_rcnn_inception	0.2	1	18321	150MB	0.9

According to Figure 14, after successful training of the CNN models, the recognition system evaluates and can determine riders with and without helmets.

[FIGURE 14: Image Classification Output — two annotated photos: left labeled "without helmet: 0.5719, with helmet: 0.4281"; right labeled "with helmet: 0.9999"]

Figure 14. Image Classification Output

Source: https://tinyurl.com/4hkjxktr

Researchers categorized their results with labels, which can be found in Figure 14. The model can determine whether the annotated object is the rider with a helmet.

A similar approach but a different model and algorithm is used by researchers (Giron et al., 2020) as shown in figure 15. Their research uses YOLO (You Only Look Once) version 8 under YOLO models. Researchers created a custom dataset to configure the detection of the model to resolve the problem of the researchers' study.

[FIGURE 15: Object Detection Output in Video — a video frame with motorcyclist and green bounding box labeled "rider w/ helmet: 80%"]

Figure 15. Object Detection Output in Video

Source: https://tinyurl.com/4hkjxktr

Researchers validate and test their models, and after a satisfactory result, they deploy their recognition system using Roboflow UI, which can be seen in Figure 16.

[FIGURE 16: Deployed Model Hosted by Roboflow UI (PC Version) — screenshot of the Roboflow web interface with sample images, upload areas, and detection output panels]

Figure 16. Deployed Model Hosted by Roboflow UI (PC Version)

Source: https://tinyurl.com/38aw3xah

According to the researchers, one of the weaknesses of the model is that it is trying to detect a fast-moving object in real-time scenarios. But YOLOv8 is effective and performs well. Researchers report that the precision of the model stayed around 0.85-0.90, which indicates a

balanced performance in terms of reliability and recognition of helmets. Susa et al. (2022) use another YOLO model. According to the researchers, it is a fast and accurate detection model. Researchers are trying to make a system that can recognize helmets that are authorized to be used; an example of this is Figure 17.

[FIGURE 17: Image frame testing result — two side-by-side photos: an annotated dark sci-fi style helmet, and a photo of a man wearing a black motorcycle helmet]

Figure 17. Image frame testing result
Source: https://tinyurl.com/56smsw5c

Heavy training of the model is needed to get the desired results and increase its accuracy. Based on Figure 18 and Figure 19, researchers' training and validation of its loss decreases on later layers.

[FIGURE 18: Training and Validation — line chart showing Training Loss (blue) and Validation Loss (red) over Epochs 1-25, starting near 40 at epoch 1 and decreasing rapidly to near 0 by epoch 8]

Figure 18. Training and validation result
Source: https://tinyurl.com/56smsw5c

[FIGURE 19: Training and Validation — line chart showing six loss curves over Epochs 1-25 (Val Layer 1-3 Loss, Layer 1-3 Loss); all curves decrease rapidly from high values to near 0 by epoch 5-8]

Figure 19. YOLOv3 layers' result
Source: https://tinyurl.com/56smsw5c

After training and model evaluation, researchers then make a helmet detection application, which is an application that you can use with three (3) different features: image detection, video detection, and live streaming.

[FIGURE 20: Helmet detection system application — GUI with title "HELMET DETECTION SYSTEM", a helmet image, and three Process buttons: Import Image, Import Video, Live Detection (subtitle: An Efficient Safety and Authorized Helmet Detection using Deep Learning Approach)]

Figure 20. Helmet detection system application
Source: https://tinyurl.com/56smsw5c

In the application made by the researchers, the three (3) mentioned features can be seen in Figure 20. You can upload images for the system to detect a helmet; another is the import of

video where you can upload a video, and the system will create a CSV file for each frame that it detects, and lastly, Live Detection, which uses live footage from a video device that can be seen at Figure 21.

[FIGURE 21: Live stream video testing result — two annotated images: one labeled "Unauthorized Helmet: 99.96%" (person with sunglasses, white helmet); another labeled "Authorized Helmet" (black motorcycle helmet)]

Figure 21. Live stream video testing result

Source: https://tinyurl.com/56smsw5c

In this figure is the sample image wherein the researcher's system performs its task, design to recognize which person uses an authorized helmet or not.

[FIGURE 22: Steps of Helmet Detection — vertical flowchart: receive input image -> convert RGB to gray scale -> background subtraction -> threshold method -> mathematical morphology method -> vehicle classification -> region of interest for motorcycle -> helmet detection]

Figure 22. Steps of Helmet Detection

Source: https://tinyurl.com/296ujrzj

Swapna et al. (2019) described helmet detection as capturing and analyzing road images or video. As shown in Figure 22, the process starts with image acquisition and preprocessing, including grayscale conversion, background subtraction, and noise removal. Vehicles are classified by size and shape to identify motorcyclists. The rider's head is extracted, and a machine learning classifier determines whether a helmet is worn.

[FIGURE 23: Image Preprocessing Steps for Helmet Detection — three side-by-side images: Original image of head (color photo), After conversion to binary (luminance threshold = 0.98), After preprocessing]

Figure 23. Image Preprocessing Steps for Helmet Detection

Source: https://tinyurl.com/ykhepn4v

Image preprocessing involves converting the original head image to a binary format, followed by several noise reduction techniques as can be seen in figure 23.

[FIGURE 24: Demarcation of Region of Interest (ROI) — circular head silhouette divided into four quadrants (Q1, Q2, Q3, Q4) with center point O and radius r]

Figure 24. Demarcation of Region of Interest (ROI)

Source: https://tinyurl.com/ykhepn4v

As shown in Figure 24, the researchers focus on the top 25% of the motorcycle blob, which is assumed to be the head area. This part is separated for more analysis and divided into four quadrants. The researchers extracted the features from these quadrants, like the circular shapes, average brightness, and colors.

[FIGURE 25: ROC Curves of different classifiers — four ROC subplots showing True Positive Rate vs False Positive Rate: Logistic Regression (AUC = 0.9049), Multi-Layer Perceptron (AUC = 0.8549), Support Vector Machines (AUC = 0.9243), Weighted Average (AUC = 0.9176)]

Figure 25. ROC Curves of different classifiers for detecting helmets

Source: https://tinyurl.com/ykhepn4v

Figure 25 shows the ROC curves for the different classifiers tested by the researchers in this study. The results indicate that the combined approach of logistic regression, multi-layer perception, and SVM classifiers achieve the best performance, with an AUC of 0.99. This indicates excellent helmet detection, with 96% accuracy and only a 2% error rate.

Related Patents

The researchers reviewed five relevant patents, as outlined in Table 3. The selected and enhanced components of the system were thoughtfully identified and incorporated by the researchers, contributing to the development of the proposed prototype.

Table 3

Related Patents

Patent Title	Author	Feature Adopted	Enhanced Feature
Method and apparatus detecting a target	Xu et al. (2018)	Deep Learning with Convolutional Neural Networks (CNNs)	Implementation of YOLOv8 for Enhanced Detection
A helmet detection method with lightweight backbone based on yolov3 network	Chang et al. (2020)	Helmet Detection Using YOLOv3	Real-Time Detection and Analysis with YOLOv8
Safety helmet wearing detection and identification method and system based on video face image	Chai et al. (2021)	Video Frame Analysis for Helmet Detection	Real-Time Detection Using YOLOv8
Helmet wearing detection method and device	Tian et al. (2024)	Helmet-Wearing State Detection	Real-Time Video Analysis Using YOLOv8

Helmet wearing determination method, helmet wearing determination system, helmet wearing determination apparatus, and program	Takahashi (2016)	Image Processing for Helmet Detection	Real-Time Video Processing with CCTV Camera

As illustrated in Figure 26, the patent “Method and apparatus detecting a target” by Xu et al. (2018) describes a sophisticated approach to object detection using a cascaded convolutional neural network (CNN). This design incorporates multiple CNN stages, each connected by thresholding layers, which progressively refine the detection process. This step-by-step filtering strategy improves overall accuracy of identifying genuine targets while reducing false positives.

This multi-level system optimally balances false positive rates (FPR) and true positive rates (TPR), increasing accuracy by iteratively training and testing at every level. The multi-stage technique reduces false alarms by sequentially eliminating erroneous results.

[FIGURE 26: Cascaded CNN flowchart — Input image -> CNN 1 -> Threshold determining 1 -> (Yes) CNN 2 -> Threshold determining 2 -> ... -> Yes -> True face; (No paths -> False face). Reference numbers 210, 215, 220, 225, 200.]

Figure 26. Flowchart of the Cascaded Convolutional Neural Network (CNN) for Object Detection

Source: https://patents.google.com/patent/US20180157899A1

As shown in Figure 27, the focus patent “A helmet detection method” by Chang et al. (2020) suggests light-weight helmet detection with the help of the YOLOv3 network along with light-weight backbone. YOLOv3 model adopts a backbone network called Darknet-53 and processes images on several layers. The multi-layer process allows the system to detect objects of different sizes at different distances with the image, with the compromise between speed and accuracy. That makes YOLOv3 especially suited to real-time tasks, e.g., industrial or construction site safety monitoring.

The patent also calls for using a light backbone, which reduces the computational load, making the system more suitable for deployment on systems with weak processors. Drawing from YOLOv3's strength, the approach provides a scalable, reliable solution to helmet conformity in various conditions.

[FIGURE 27: YOLOv3-Based Helmet Detection System architecture — IMAGE (416 x 416 x 3) -> Darknet-53 -> three branches outputting Y2 (26 x 26 x 255), Y1 (13 x 13 x 255), Y2 (52 x 52 x 255), with DBL & Up-sample blocks and Concat operations]

Figure 27. Architecture of YOLOv3-Based Helmet Detection System with Lightweight Backbone
Source: https://patents.google.com/patent/AU2020100705A4

As seen from Figure 28, Chai et al.'s (2021) “Safety helmet wearing detection and identification method and system based on video face image” is a comprehensive and systematic approach to helmet detection using video frames. The initial step is video image capture, which is face detected and localized to identify individuals in the video stream. The systematic approach is based on a series of crucial steps, namely image processing, feature extraction, and recognition, which support one another to be highly precise in determining whether individuals wear helmets.

To enhance the facial sharpness of detected faces, the video frames are filtered and smoothened during the image processing process. After face detection, feature extraction is performed to collect data of relevance concerning the appearance of the subject and whether a helmet is worn. In determining whether a helmet is being worn or not, the recognition process subsequently checks the obtained attributes against a predefined database. Even during bad weather, our approach consistently distinguishes riders with helmets from others.

[FIGURE 28: Video-Based Helmet Detection System block diagram — vertical flowchart: Video Image Capture -> Face Detection and Localization -> Image Processing, Feature Extraction, and Recognition -> Foreground Detection -> Human Body Detection and Localization -> Head Region Localization]

Figure 28. Block Diagram of the Video-Based Helmet Detection System Using Face Imaging Processing
Source: https://tinyurl.com/2ncxx82n

As depicted in Figure 29, the patent “Helmet wearing detection method and device” by Tian et al. (2024) presents a method for detecting helmet use. This method involves a sequence of operations, beginning with detecting an unlocking command that unlocks both the vehicle and the helmet lock and displays a prompt message. Upon removing the helmet from the lock, data is collected from the RFID tag embedded in the helmet, enabling real-time monitoring of helmet use. This patent demonstrates that a motorcycle's electronic systems can be integrated with

helmet use, enhancing safety through technological monitoring. The system can also alert authorities or users if the helmet is not worn after the vehicle is unlocked.

[FIGURE 29: Operational Flow of Helmet Wearing Detection — flowchart steps S1-S4: S1: When the unlocking command is detected, execute the vehicle unlocking and helmet lock unlocking command, and play a prompt message. S2: After detecting that the helmet is removed from the helmet lock, collect data from the RFID tag placed inside the helmet. S3: Use the RFID reader inside the helmet compartment to receive the data and monitor the real-time helmet-wearing status based on the helmet-wearing detection model. S4: After detecting that the helmet is removed from the helmet lock, collect data from the RFID tag placed inside the helmet.]

Figure 29. Operational Flow of Helmet Wearing Detection and RFID-Based Monitoring System
Source: https://patents.google.com/patent/CN117953614A

Finally, as shown in Figure 30, Takahashi (2016) shows a conceptual block diagram in their patent entitled “Helmet wearing determination method, helmet wearing determination system, helmet wearing determination apparatus, and program,” showing the structure of their helmet detection system. This diagram visually represents the interactions between the various system components, highlighting the crucial role of image processing in determining helmet use, and shows the system's potential to monitor traffic in real time by allowing continuous observation, and shows the system's potential for real-time traffic monitoring by enabling continuous tracking, quick identification of violations, and immediate reporting to authorities for faster enforcement.

[FIGURE 30: Conceptual Block Diagram — IMAGING DEVICE 1 -> IMAGE -> HELMET WEARING DETERMINATION APPARATUS 2 -> DETERMINATION RESULT]

Figure 30. Conceptual Block Diagram of the Helmet Wearing Determination System
Source: https://patents.google.com/patent/US20160292520A1

In addition, Figure 31 shows the best position for installing the imaging device, which is essential for successful helmet detection. This figure shows how the imaging device should be located to record relevant visual information of the rider. Correct camera location is crucial to effective monitoring and detection of helmet wearing, hence optimizing the entire effectiveness of the system. Both images combined demonstrate an overall realization of helmet-wearing detection technology's theoretical and practical concepts.

[FIGURE 31: Imaging Device Installation — placement of multiple cameras around a roadway with motorcycle traveling, indicating "TRAVELING DIRECTION" arrows and convergence at IMAGING DEVICE 1 position]

Figure 31. Imaging Device Installation for Effective Helmet Detection

Source: https://patents.google.com/patent/US20160292520A1

Proposed Solutions

Considering the underlying reasons for the issue, the researchers implemented the following software and hardware fixes into the design and creation of the project study:

Table 4

Proposed Solution Matrix

Problem	**Root cause**		**Proposed Solution Feature**	
			Hardware	**Software**
Motorcycle riders continue to disregard regulations and law by riding without helmet	Materials	Traditional Video Surveillance (Aquitan et al. 2023)	Helmet recognition system overlay on monitoring system	Python, Anaconda Prompt, JavaScript, Visual Studio Code, Webhost
	Measurement	Multiple Checkpoints (Susa et al. 2022)		

In solving the problem stated in Table 4, the researchers chose two root causes and suggested hardware and software solutions suitable for its current circumstances. The researchers created a helmet detection system from one of the YOLO models, YOLOv8, as a countermeasure to wipe out the cause: conventional video surveillance and some checkpoints of the police utilizing the solution. The system utilizes real-time image processing to automatically detect if a motorcyclist wears a helmet, minimizing the amount of human interaction and maximizing efficiency. Through automation the helmet detection process, the system makes it easier for law enforcement to monitor traffic better, with enhanced adherence to safety standards. Besides, the application of YOLOv8 integrates to make the system detect at high speeds precisely, making it fit for dynamic traffic conditions.

Conceptual Model of the Study

This research brought to the fore a holistic approach of enhancing road safety by enhancing helmet detection through the incorporation of high-definition CCTV cameras. As

motorcycle accidents continue to be of serious concern, having riders obey safety measures, such as donning helmets, is a priority in limiting injuries and death. The framework concentrates on merging advanced image processing technologies and high-definition video captures by CCTV cameras to make real-time tracking of use of helmets by motorcyclists. By utilizing these technologies, the system can provide immediate alerts to authorities when a rider is detected without a helmet, allowing for timely enforcement of safety regulations. This model is divided into three (3) parts which is Input, Process and Output which can be seen in the figure 32.

Input	Process	Output
Knowledge Requirements • Helmet Use and Safety Enforcement • Process Creation of Real-time Automatic Recognition of Motorcyclist with or without helmet • Technology Integration in Helmet Recognition • Advance Image Processing Technique • Data Compilation and Experimental Setup • YOLO (You Only Look Once) workaround process **Hardware Requirements** • System Unit • CCTV Camera • Ample Storage for datasets, Outputs and Programs • Sufficient length of Cables **Software** • Operating System: Windows • Compilers and Libraries • Additional Tools for Video Processing	**Design** • Analysis, calculation and design of the system and prototype **Pre-processing** • Image Annotation • Dataset Training • Web-based Database **Feature Extraction and Classification** • Motorcycle Recognition • Helmet Recognition • Data Analysis **Algorithm Testing** • Testing Models • Data Extraction **Evaluation Metrics** • Functional Suitability • Performance Efficiency • Compatibility • Usability • Reliability **Revision and Modification Implementation**	**Output** Helmet Recognition System with Algorithm Analysis

Figure 32. Conceptual Model of the Study

Input

The Data requirement was reviewed to find the most efficient methods to test the prototype's performance. A helmet recognition system must meet several hardware, software, and knowledge requirements to be successfully developed. Understand helmet use and safety rules, explore real-time automatic recognition system for motorcycle riders putting on or off, interface for helmet recognition using technology, and apply advanced image processing techniques. Data compilation techniques and background subtraction techniques are also used to help identify moving vehicles. It's also important to be knowledgeable about the YOLO (You Only Look Once) workaround process. Research shows that implementing traffic enforcement, such as helmet usage, significantly reduces the number of motorcycle accident fatalities and brain damage. The hardware requirements include a system unit, CCTV cameras, enough wiring, and enough storage for datasets and outputs. Windows OS and Visual Studio Code as the primary IDE, which consists of fundamental libraries and tools such as Python, Visual Studio, and the Anaconda distribution for software development, while other libraries such as OpenCV for image processing and NumPy for numerical operations are integrated to allow real-time image analysis and data processing. These elements will ultimately put the helmet recognition system in place for improved road safety.

Process

The process involves pre-processing steps like image labeling, generation of training datasets and web-based database. Feature extraction and classifying process on completing the research project, algorithms identify helmet-wearers and motorbikes. Models learn data during this phase using labeled datasets, and performance is checked by testing them with different training models. This looping enhances the algorithms to become more accurate in real world applications. The process ensures continuous system improvement, enhancing helmet detection, reliability and performance.

Output

The product of this research was a helmet recognition system developed with algorithm analysis. The system should be capable of identifying motorcyclists whether they wear helmets or not while riding. Once the system detects whether one wears a helmet or not it is classified in the web-based database.

Operational Definition of Terms

This part of the chapter presents a glossary of terms utilized in the study and their meanings to assist in understanding the research, which are as follows:

Algorithms are procedures carried out step by step or equations employed in problem-solving, especially in computer programming and data processing.

Artificial Intelligence is a field of computer science that tries to build systems that can execute tasks normally requiring human intelligence, for example, the comprehension of language, the identification of images, or decision-making.

Automate is an activity of applying technology in the absence of human intervention.

Background Subtraction is a method of image processing in which moving objects are extracted from the background of a series of video frames.

CCTV which stands for Closed-Circuit Television, is an electronic surveillance system in which signals are delivered to designated monitors for monitoring and security purposes.

CNNs refer to a category of deep learning algorithms used mostly in image processing tasks, that process visual data by feeding it through several layers of detectors of different features.

Dataset is a set of organized data employed for analysis, model training, or other research intents.

Detection Systems are technologies designed to identify specific objects, patterns, or activities and are used predominantly in security and automation.

False Positive Rates (FPR) are the rate of false positive identification, where a system incorrectly recognizes an object or event that does not exist.

FasterRCNN is a deep learning model that has been specially tuned for object detection tasks, is renowned for being fast and accurate in recognizing objects in images or video.

Grayscale Conversion refers to the process of converting a color image to shades of gray and is commonly used in image processing to reduce data complexity.

Helmet is a form of protective wear that motorcycle operators can wear in order to gain safety and reduce possible injuries in cases of accidents.

Detection of Images is the capability of computer vision to recognize features or objects based on images.

Image processing is manipulating or analyzing algorithms of digital images, mainly aiming at improving them or extracting meaningful information from those images.

Inceptionv2 is a designed convolutional network model to advance object detection along with the optimization of image recognition efficiency.

Live streaming refers to the live, simultaneous broadcast of video content over the internet or a network.

Machine learning is an artificial intelligence discipline that is centered around training algorithms on patterns so that they can make real decisions based on data, not explicit programming.

Metropolitan Manila Development Authority (MMDA) is a government agency in the Philippines responsible for managing and overseeing development and public safety in the National Capital Region.

MobileNetv2 is a lightweight deep-learning model optimized for mobile and embedded vision applications, particularly for tasks requiring speed and efficiency.

Monitoring is the continuous observation or tracking of a process, system, or activity, typically for management or safety purposes.

Motorcycles are two-wheeled motor vehicles commonly used for personal transportation.

Noise Removal is an image processing technique that filters out unwanted artifacts or distortions from images.

Offline System is a system or application that operates without a connection to the internet or network, performing tasks locally on a device.

Power over Ethernet (PoE) is a technology that enables Ethernet cables to deliver both data and electrical power to network devices, such as IP cameras or wireless access points, through a single cable.

Prototype is an early model or version of a product or system used to test concepts and functionality before final development.

Real-time Applications are systems or applications that respond instantly to inputs or changes, often used in critical scenarios like monitoring or automation.

RFID also called Radio Frequency Identification, is a technology that uses electromagnetic fields to identify and track tags attached to objects.

Roboflow UI is the user interface tool for managing datasets, training, and deploying computer vision models, typically used in image detection projects.

Surveillance is the continuous observation of activities, typically for security, law enforcement, or monitoring purposes.

Traditional Video Surveillance is a standard method of monitoring environments using CCTV cameras to record and observe activities over time.

True Positive Rates (TPR) are the proportion of correct positive identifications where a system accurately detects the presence of an object or event.

Vehicles are any form of transportation designed to carry passengers or goods, including cars, motorcycles, and trucks.

Video Footage or recorded video content is often used as evidence or for review in monitoring systems.

YOLO (You Only Look Once) is a real-time object detection model known for processing images in a single pass, achieving fast and accurate detection results.

Chapter 3

METHODOLOGY

This chapter includes the project design of the system, such as the drawing projections, block diagrams, and process flow diagrams with the following concept of the study. These figures have discussions to aid in the development of the proposed system. It also presents the system's project development, such as the working drawings, interface design, and step-by-step procedures of the prototype fabrication and the creation of the whole system for data analytics of the study. The operational, testing, and evaluation procedures are also presented in this chapter.

Project Design

[FIGURE 33: Architectural Diagram of the System — shows physical layout: a motorcycle with helmeted riders feeds Video Footage Recorded into a CCTV camera; the camera connects via LAN Cable through a PoE switch (LAN Cable) to a desktop computer running the YOLOv8 Model, which Annotates Footage and feeds a Web-based System Dashboard and Stores Video Footage and Annotated Images.]

Figure 33. Architectural Diagram of the System

The project design highlights essential components, structure, and processes required to achieve the study's objectives. It includes hardware and software integrations, block and flow diagrams, and the sequential operations necessary for the system's functionality.

The proposed Helmet Recognition System prototype should auto-detect those motorcycle riders putting on and off helmets while going through a place. With this, it employed a high-definition CCTV camera wherein real-time videos are being used, then applied the YOLOv8 object detection algorithm; thus, able to classify that the rider with the motorcycle on the road wore or didn't wear the helmet accurately wherein all these processes done inside the computer as seen in Figure 33.

The prototype includes the following components:

1. Input: Live video footage of motorcycles recorded by a CCTV camera.

2. Processing using YOLOv8 algorithm to analyze real-time video from riders with or without helmets.

3. Storage: Results of detection in a database for further analysis and report.

4. User Interface: The real-time detections are shown on a web-based dashboard with summarized data and analytics for the authorities.

The system will work in real-time, so it will not only help local authorities with insights to enforce helmet laws but also minimize motorcycle-related accidents. By providing immediate feedback on helmet usage, the system promotes better road safety awareness among riders. This proactive approach contributes to creating safer roads, reducing injuries, and enhancing compliance with safety regulations.

Block Diagram

The block diagram on Figure 34 was made through the Lucid.app, which demonstrates that YOLOv8 and a camera can make storing data and helmet recognition easier. The camera records a live feed video after initializing the system, which, in real-time, YOLOv8 analyses to figure out who has helmets on; shows results in real-time; and allows users to view settings, monitor checkpoint activity and save photos, as well as access data. The main processing unit decides since it stores relevant information regarding headgear and unhedged motorbike riders, including images related thereto for examination or storage, and through an integration of this hardware (unit for processing with camera) along with software known as YOLOv8, thereby ensuring that safe data storage happens, while efficiency in helmet identification helps in observation of safety provisions at checkpoints.

Figure 34. System Flow of the Helmet Recognition System

Flow Chart

Figure 35 represents the helmet recognition and data management process, developed using Lucid.app. It starts with powering on and camera initialization. The system captures live video feed processing in real time using YOLOv8 for helmet detection. The system then reaches a decision point of whether a helmet is detected or not. The system logs the rider as complaint if a helmet is detected, saves the image and data for further analysis, and updates the statistical records. In case no helmet is detected, the system logs the rider as non-compliant, captures and stores the image.

The processed results are displayed on the user interface, allowing for real-time monitoring, configuration, and interaction. The system captures and analyzes the new feeds of video into it continuously such that the helmet continues to be recognized rightly to collect data on the enforcement of road safety. The process is strictly sequential such that the system is first powered and the CCTV camera activates, records live video and forwards the video for processing. YOLOv8 starts to analyze the video to identify riders and helmets. The system then checks for the presence of a helmet. If the helmet is detected, the system logs the rider as compliant, saves the picture and updates the statistical report. If no helmet is found, then the system logs non-compliance, saves the picture, alerts the authorities (if that feature is enabled) and updates the statistical report.

[FIGURE 35: Flow Chart of the Helmet Recognition System — vertical flowchart: System Initialization -> Capture Video Feed -> Detect Motorcycles & Helmets -> [decision diamond] Helmet Detected? (No -> loop back to Capture Video Feed; Yes -> Store Data) -> Store Processed Data -> loop back to Capture Video Feed.]

Figure 35. Flow Chart of the Helmet Recognition System

Description and Discussion

The system uses a single camera installed in front of a checkpoint to provide feed video for helmet recognition. YOLOv8 is a state-of-the-art real-time object recognition algorithm known because of its accuracy and speed, which is used for processing video feeds.

The camera captures the video, and the YOLOv8 algorithm processes it in real-time, allowing helmets to be detected around. When the system identifies a helmet, it records

information about the rider with the helmet and saves the taken images to analyze and authenticate in the future. When no helmet is sensed, the system gathers data regarding the non-helmeted rider and saves the corresponding images. The system architecture allows continuous video feed analysis and real-time recognition, with results presented on a user interface for easy inspection and user interaction.

This setting ensures that the most precise and trustworthy identification technique is used, thereby providing an efficient solution for helmet identification during inspections at check points.

For a better idea of how the system operates and what it comprises a flowchart was developed in Lucid.app. The flowchart describes the step-by-step process from video capture to helmet identification and data retrieval. A block diagram, also created in Lucid.app, for instance, gives a detailed description of the hardware and software elements of the system and how the camera, YOLOv8 algorithm, data storage, and user interface work within the system.

Project Development

In this project research, the researchers employed the Waterfall SDLC model, as indicated in Figure 36 by Alshamrani and Battab (2015), which illustrates the stages: Analysis, Design, Development, Testing, Implementation, and Maintenance.

[FIGURE 36: Waterfall Model Life Cycle — diagonal cascade of six labeled boxes flowing top-left to bottom-right: Analysis -> Design -> Development -> Testing -> Implementation -> Maintenance, each connected by a downward-then-rightward arrow.]

Figure 36. Waterfall Model Life Cycle

Source: https://tinyurl.com/mtawwmtm

Analysis Phase

This stage discussed the experts' evaluations on current technologies and consulted different stakeholders to evaluate system requirements. It was noted that some particular applications, for instance, YOLOv8 and Python, and particular hardware, such as a PC and CCTV camera, were needed. Besides, the stage emphasized the necessity of providing attention

to the system's scalability and flexibility to multiple environments for smooth deployment.

Design Phase

During the Design phase, system architecture was established at the component level, explaining how PC and CCTV camera communicate at hardware component level and how their software equivalents function in YOLOv8 and Python.

Development Phase

It was developed using YOLOv8-based detection software within a local Anaconda environment named “yolov8_custom” within an under-equipped system with an NVIDIA GeForce GTX 1060 3GB GPU. It was trained first using 250 datasets from OpenImages and improved by 1648 custom images annotated through CVAT.ai for three classes of motorcycle, helmet, and head. The training was repeated with different datasets, and the accuracy kept increasing after adding 353 more images. The GPU posed a challenge because of the lengthy extended periods required for the training process, up to 18 hours per session. The trained model was then uploaded onto a local server and web-integrated in a way that can facilitate real-time data visualization using HTML, PHP, and MySQL.

Figure 37 shows the Anaconda Prompt, which utilizes a Command Prompt interface specifically designed for Python-related tasks. It plays a crucial role in running the script created in Visual Studio, enabling the execution of the detection system. This tool allows seamless integration between the development environment and the Python execution process, ensuring smooth operation of the helmet recognition system.

[FIGURE 37: Anaconda Prompt of the System — screenshot of a dark Anaconda Prompt terminal window showing the command line: (yolov8_custom) D:\DETECT>python predict.py]

Figure 37. Anaconda Prompt of the System

The Anaconda Prompt was used to run the Python scripts, which ensured correct connectivity to load the IP address of the CCTV camera, which was connected via a PoE switch for the live footage annotation. However, the system could also process recorded videos and

images. From Figure 38 the Anaconda Prompt was also the interface by which the user would deploy the detection script to the correct location in the files and execute some commands.

[FIGURE 38: Anaconda Prompt executing commands — screenshot of an Anaconda Prompt terminal showing a sequence of commands: (base) C:\Users\YOLO>d: (base) D:\>cd detect (base) D:\DETECT>conda activate yolov8_custom (yolov8_custom) D:\DETECT> python predict.py]

Figure 38. Anaconda Prompt executing commands

Figure 39 captures YOLOv8, that is, the main software and algorithm responsible for processing and analyzing the data that the CCTV system takes. It is accessible at https://docs.ultralytics.com/models/yolov8/ and installed on the PC as it begins to run simultaneously with the system upon its activation.

[FIGURE 39: YOLOv8 Object Detection Algorithm Execution — screenshot of the Ultralytics YOLOv8 documentation page (docs.ultralytics.com) showing the "Ultralytics YOLOv8" overview heading, navigation sidebar listing YOLO model versions (YOLOv3 through YOLOv11, plus SAM, MobileSAM, FastSAM, YOLO-NAS, RTDETR, YOLO World), an overview paragraph, and two performance comparison charts plotting COCO mAP vs Parameters (M) and Latency A100 TensorRT FP16 (ms/img) for the different YOLO versions.]

Figure 39. YOLOv8 Object Detection Algorithm Execution

Testing Phase

This phase was a rigid testing phase wherein the system had to be of the required standard. Tests concentrated on the helmet detection accuracy and different lighting conditions. The actual data were taken to check the system's performance, specifically detection speed and accuracy. Other problems, such as wrong detection for helmets, were then adjusted with appropriate model parameters by further training them with additional data.

Implementation Phase

The system was installed in the chosen location for live monitoring after successful testing. Local authorities were equipped to operate the system, interpret the results, and utilize insights for road safety improvements. The development team arrived at a decision on the resolution of installation challenges, such as network configuration and integration with existing infrastructure and system installation.

Maintenance Phase

To ensure long-term reliability, regular system maintenance routines were established. These routines involved regular checks on hardware components, software updates, and model retraining to accommodate new data.

Fabrication Procedure

The researchers followed the step-by-step installation process of the Helmet Recognition individual modules. The following sections discuss and present the prototype of the system, HRS.

Camera Installation

- Unpack the CCTV cameras and check if all components are complete and not damaged.
- Confirm that each camera comes with the required mounting hardware and data cables.
- Position the CCTV cameras for optimal traffic coverage at the designated locations.
- Ensure that the cameras have a clear and unobstructed view of the monitoring area.
- Use the mounting equipment provided to secure each camera in place.
- Connect the camera to the central processing unit using the network cable, particularly the LAN cable, to a switch with the Power on Ethernet feature.
- Ensure that the connections are secure and there are no loose ends.
- Verify that each camera is receiving data and power and is operational.

Processing/System Unit Setup

- Ensure the system unit is plugged in and receiving power from an outlet.
- Install the YOLO algorithm and necessary software into the system unit.
- Ensure that the algorithms are correctly uploaded and configured.
- Connect the processing unit to the cameras using the data cable and connect to the local area network (LAN).
- Test the connection if the CCTV's IP Address is detectable in the network.

- Double-check that all connections (data and power) are tight and secure if the connection is not made.

System Integration

- Install the necessary software on the system unit.
- Verify that the software is compatible with the YOLO algorithms.
- Upload and configure the YOLO algorithms within the software.
- Ensure that the algorithm is set up to process video feeds from the respective cameras.
- Verify that the system unit and software work properly and work as intended.

System Calibration

- Conduct initial tests to calibrate camera focus and algorithm performance.
- Adjust the camera angles and settings to optimize video capture quality.
- Fine-tune algorithm settings based on preliminary recognition results for optimal performance and accuracy.
- Ensure that the system correctly identifies helmet usage with minimal errors.

Connecting the System to the Display Interface

- Plug the proprietary power cord of the display or monitor into the main power source.
- Gather the necessary connecting cables.
- Verify that the cables are intact and free of damage.
- Connect the system unit to the display interface using the connecting cables.
- Ensure that all connections are securely fitted to prevent any disconnections.
- Test and calibrate the display and ensure it is operational, and all components are functioning correctly.

Operation and Testing Procedures

These testing procedures aimed to determine if the researchers' prototype was fully functional and assess the functionality of the hardware and software components. The testing also helped identify any potential issues or areas for improvement, ensuring the system's reliability

and effectiveness in real-world applications.

1. A camera is mounted in an optimal place to guarantee its detection and recognition of motorcycle riders not wearing a proper helmet while preventing unwanted incidents.

2. The utilization of YOLOv8 detects motorcycle riders in captured images or video frames. The data captured by the camera will be stored in the computer local storage.

3. A data analytics application based on object detection was constructed using Python language and was utilized to analyze the data, counting number of riders without helmets. This encompasses summarizing and interpreting the output from the object detection and helmet recognition stages.

Testing Concepts and Procedures

An exhaustive test program was adopted to verify the helmet recognition system was correct and trustworthy. Testing was performed on such parameters as the criticality of real-world functionality, such as correctness, precision, and recall of the system in recognition.

Hardware Testing

Camera System. The device requires a high-mount and facing the flow of the traffic on the camera system to be properly able to identify and recognize the motorcycle riders and the condition of the helmet. Testing procedures. Installation involved placing the camera system at a favorable height and calibrating it for optimal detection and recognition capabilities.

Computer System. The device does all the hard work, like processing the recognition system and storing all collected data required for data analytics.

Software Testing

YOLOv8 Integration. Python was used to merged the YOLOv8 in the system for effective object detection and helmet recognition operations. The YOLOv8 model was trained on multiple datasets to refine its performance on detecting motorcycle drivers and detecting helmet wearing. Python further offered the support to personalize and fine-tune the YOLOv8 model so that real-time processing and examination of video feeds, the accurate and swift helmet detection being imperative in numerous environments.

Web-Based UI. Local web host where the result can be seen, such as the summary of the detection and the live footage of the Camera system. Website that is programmed using HTML, CSS, JavaScript, and PHP for its design and databases.

System Functional Testing

It was performed to check the features and functionalities of the system to ensure that the hardware and software components perform the expected output. The following steps were undertaken for each iteration.

1. Define and distinguish the core functions of the helmet recognition system prototype, focusing on helmet detection using YOLOv8, and database storage of captured data.

2. Create diverse scenarios using motorcycle riders, some wearing helmets and others not, to simulate real-world conditions such as different lighting and environmental factors.

3. Execute predefined test cases to validate each function of the system, including accurate helmet detection.

4. Compare the expected outcomes with the actual results obtained during testing to identify any discrepancies.

5. Analyze discrepancies between expected and actual results to pinpoint potential issues or areas for enhancement in the system's performance.

6. Establish specific pass/fail criteria based on system requirements and performance to assess overall functionality.

Evaluation Procedure

To gather all necessary information and to get the performance result of the project, a series of tests and surveys were conducted. This study used the ISO 25010:2011 evaluation instrument for software. The system was evaluated based on the capabilities the system is intended to perform. The evaluation of the system was based on its ability to execute planned features.

The general requirements for the evaluation were the following: (1) functional suitability, (2) performance efficiency, (3) compatibility, (4) usability, (5) reliability, and (6) portability of the output system.

The following steps were used during the evaluation process:

1. *System Presentation*

   The system was presented to the LGU of the chosen barangay.

2. *System Examination and Navigation*

   The system can be experienced by both end-users and administrators who will provide feedback.

3. *Survey Distribution*

   The survey questionnaires were distributed to get feedback from the system. The survey questions were based on the principles of ISO 25010:2011, and this system was rated using the Likert Scale, as shown in Table 5.

Table 5
Likert Scale and Interpretation

Numerical Scale	**Descriptive Reading**
5	Excellent/Highly Acceptable
4	Very Good/Very Acceptable
3	Good/Acceptable
2	Fair/Fairly Acceptable
1	Poor/Not Acceptable

Based on the results obtained for each category, Table 6 presents the evaluation range and corresponding interpretation of the numerical scale. This table serves to interpret the reliability of the gathered data.

Table 6

Evaluation Range and Interpretation

Numerical Scale	Descriptive Reading
4.51 – 5.00	Excellent
3.51 – 4.50	Very Good
2.51 – 3.50	Good
1.51 – 2.50	Fair
1.00 – 1.50	Poor

4. *Survey Validation*

To procure the required data and measure the performance of the project, a series of tests, in addition to the responses of the respondents, is examined together with the results and respective calculations. Based on this analysis, a thorough idea about the Helmet Recognition System is obtained, gaining an understanding of its performance, users' satisfaction, and effectiveness. The results give an evaluation of the degree to which the system accomplishes its objectives and identify areas of potential improvement, according to the responses' assessment.

Evaluation Tool for the system that is located in page 128, Appendix D. It presents an all-encompassing framework for determining the system's performance and efficacy on various levels, such as functional suitability, performance efficiency, compatibility, usability, reliability, and portability. Survey participants will evaluate each criterion using a Likert scale, making it possible to have an orderly and measurable examination of the effectiveness of the system in fulfilling its goals. This method provides for the assessment of the HRS based on both technical performance and user satisfaction.

[Editor's note: Page 71 of the original thesis was not photographed in the upload and is therefore not reproduced here. The narrative continues with Table 7 below.]

Table 7
Summary of Mean Score from the Evaluation

Question Number	Mean

1	4.4
2	4.2
3	4.4
4	4.7
5	4.4
6	4.6
7	4.9
8	4.9
9	4.7
10	4.8
11	4.9
12	4.3
13	4.7
14	4.8
15	4.8
16	4.6
17	4.4
18	4.7

The system's performance was evaluated using Cronbach's Alpha, which measures the reliability and internal consistency of the dataset gathered. The evaluation was then completed by calculating Cronbach's Alpha, as illustrated in Figure 40. The descriptive rating given to the resulting Cronbach's Alpha value is displayed in Table 8.

$$\alpha = [\, n / (n - 1) \,] \times [\, 1 - (\Sigma\, s^2(y) / s^2(x)) \,]$$

Figure 40. Formula for Cronbach's Alpha

Where:

n is the number of the test item

$\Sigma\, s^2(y)$ is sum of the item variance

$s^2(x)$ is the variance of total score

Table 8
Descriptive Interpretation of the Cronbach's Alpha

Cronbach's α	**Internal Consistency**
0.90 and above	Excellent
0.80 – 0.89	Good
0.70 – 0.79	Acceptable
0.60 – 0.69	Questionable
0.50 – 0.59	Poor
Below 0.50	Unacceptable

5. *Software Evaluation*

After the training phase, the YOLOv8 model was evaluated and validated using precision, recall, and the mean average precision (mAP) metric. These metrics served as standards to ensure the model's effectiveness in real-world applications.

Precision measures the proportion of positive predictions that are confirmed ‘true,’ indicating correctly predicted helmets and motorcycles to all those that were detected. A higher precision metric indicates the lessened occurrence of false predictions. Conversely, recall is the metric that measures the ratio of true positives to the total number of actual positives present, determining the number of real helmets and motorcycles in the dataset that the YOLOv8 model was able to detect. A lower recall suggests a lower prediction of false negatives. Together, these metrics evaluate the model's accuracy and ability to detect relevant objects.

Mean Average Precision (mAP) metrics measure the YOLOv8 model's overall detection accuracy by evaluating its performance across different Intersections over Union (IoU) thresholds. The objective of this metric was to evaluate the model's ability to detect an object by analyzing the overlap between the actual coordinates of the target and the predicted bounding boxes.

$$Accuracy = (\#\ of\ Detected\ Object / Total\ \#\ of\ Objects) \times 100$$

$$Precision = True\ Positive / (True\ Positive + False\ Positive)$$

$$Recall = True\ Positive / (True\ Positive + False\ Negative)$$

Figure 41. Accuracy, Precision, and Recall Formula (Perri, Simonetti, & Gervasi, 2022)
Source: https://doi.org/10.3390/electronics11010002

As illustrated in Table 9 on page 98 the performance metrics of the trained YOLOv8 model are determined from the confusion matrix outputs as in Figure 68 page 97. These include the accuracy of 79.4%, precision of 87.5% for motorcycles and 63.6% for helmets, as well as recall which was 100% for both. This was according to the formula for True Positives, False Positives, False Negatives, and True Negatives accuracy, precision, and recall (Perri, Simonetti, & Gervasi, 2022). For the evaluation purpose, the Train18 subset of 353 datasets, trained over 30 epochs, was utilized. This requires more optimization because it has less precision in detecting a helmet.

The Performance Metrics Table is also derived after the model's training. This table is created by analyzing the confusion matrix's outputs, which include values for True Positives (TP), False Positives (FP), False Negatives (FN), and True Negatives (TN) for every class. Critical metrics like accuracy, precision, and recall are computed using these values. These metrics are essential considering that they provide in-depth information on how the model recognizes and categorizes data, such motorcycles and helmets, while reducing errors, including false positives and negatives.

Chapter 4

RESULTS AND DISCUSSION

The chapter presents the system design, which encompasses its structural elements, block diagrams, process flow diagrams, and step-by-step system development of the helmet detection system, merging the algorithm with data analytics. The operational procedures, testing methods, evaluation processes, system limitations, and future improvement recommendations are presented in this chapter.

Project Description

The helmet recognition system was developed to enhance road safety by ascertaining helmet compliance and if motorcycle riders are wearing helmets while riding a motorcycle. The system was deployed on public roads to minimize the risk of head injury and to monitor and ensure compliance.

The Helmet Recognition System was designed to enhance road safety by establishing helmet compliance and if motorcycle riders are wearing helmets while riding a motorcycle. The system was deployed on public roads to minimize the risk of head injuries and to track and enforce compliance. The system was intended to identify motorcycle helmet use by examining pre-recorded or live videos captured from roadways, intersections, highways, and particularly accident hotspots. Real-time or recorded video examination with HRS sought to enhance road safety by minimizing head injuries and crashes involving cyclists and pedestrians through effective monitoring and enforcement. The system consists of several important components: a monitor, a system unit, a CCTV camera, and a PoE switch. The CCTV camera serves to capture motorcycle riders in real-time. The video is forwarded to the system unit, where YOLOv8 processes and analyzes the video in real-time. According to YOLOv8's object detection capabilities, it recognizes whether the motorcyclist is wearing a helmet. The PoE switch provides power and data link to the CCTV camera, providing seamless and optimal operation without any

need for specific power sources. The data is transmitted to the system unit for analysis using the aid of Python programming language, which can support YOLOv8 in transforming detection outputs into data analytics. This module for analytics provides for more in-depth information, including trends in helmet compliance, such as helmet compliance patterns, reports, and other statistical information. The results are then displayed on the monitor for real-time monitoring and review.

[FIGURE 42: System Visual Output — two images: (left) close-up of CCTV camera mounted on building exterior; (right) live system feed annotated with overlaid green text “Motorcycle Count: 0, With Helmet: 0, Without Helmet: 0” plus a yellow rectangular detection zone on roadway view.]

Figure 42. System Visual Output

Figure 42 shows the CCTV camera system integrated with a YOLO (You Only Look Once) object detection framework. The camera was installed on a nearby road. The video feed captured by the CCTV is processed by the YOLO-based system to detect and track objects, specifically motorcycles. The system analyzes the video in real-time, identifying whether the motorcycle riders are wearing helmets or not.

The system counted the number of motorcycles, and it detected if a rider is wearing a helmet or not. There is a marked yellow box where the system will focus on the detection area and process the objects it finds there.

Project Structure

The system comprises six major components: the PC, CCTV camera, monitor, PoE switch, and LAN cable. In terms of software, YOLOv8, Visual Studio, Anaconda Prompt, and XAMPP are implemented for the system's operation and functionality.

[FIGURE 43: PC used in the System — photograph of the inside of the desktop PC showing CPU (cooler/fans) and GPU (graphics card) labeled with arrows.]

Figure 43. PC used in the System

As shown in Figure 43, the PC serves as the brain of the system, hosting the YOLO program, storing gathered data, and acting as the server for the web interface displaying the

results. The CPU manages system operations, such as processing video frames, handling data flow, and running the web server. Meanwhile, the GPU accelerates the computationally intensive tasks of YOLO's neural network, enabling real-time object detection and classification. Together, the CPU and GPU ensure efficient processing of the CCTV feed, identifying motorcycles and detecting helmet usage accurately.

1. Ryzen 5 3400g: This AMD processor features 4 cores and 8 threads, along with integrated graphics. With a thermal design power (TDP) of only 65W, it is an energy-efficient choice, ideal for long-term system use.

2. 16GB DDR4 RAM: This amount of memory is sufficient to run most applications without the risk of memory shortages, making it the optimal size for the program being used.

3. GTX 1060 3G Variant: Manufactured by NVIDIA, this GPU is suitable for general tasks and is fully capable of running the program without issues. It operates at 120W, making it an excellent choice for maintaining low energy costs.

4. 128GB SSD storage used for Operating system which is Windows 10 and 1TB of HDD is reserved for the YOLOv8 dataset, training and all gathered data by the recognition system.

In Figure 44 shows the monitor, which is used to display information and events processed by the PC. It allows users to observe the system's progress and real-time activities, including the live feed from the CCTV and the results gathered from the data analysis.

[FIGURE 44: Live Footage of the CCTV via Monitor — photograph of a computer monitor displaying the system's live feed interface showing CCTV view of a road scene with utility pole and trees.]

Figure 44. Live Footage of the CCTV via Monitor

Computation for the Resolution of the System (Reference for System Requirements)

Given:

1. 1 TB = 1,000,000,000 KB

2. Average size of image = 145 KB

number of images = total storage / average size of image

number of images = 1,000,000,000 / 145 ≈ 6,896,551 images

Figure 45. Number of Images per Storage Capacity Formula
Source: https://tinyurl.com/mrhhb3b6

According to Remington, G. (2023) in figure 45, the storage capacity for images depends on their average size. The calculation here assumes an image size of 145 KB. Image sizes vary based on resolution, format (JPEG, PNG, RAW), and compression. This estimate gives a rough idea of how many images can be stored on a 1 TB storage hard disk drive under these specific conditions.

Image Resolution

- Captured Image Resolution: 400×600

Video Resolution

- Input Video Resolution: 1920×1080 (Full HD)
- The System processes Full HD video at an average of 15 FPS, which is capable of handling high-quality streams with satisfactory performance. With the system unit having a total storage capacity of 1TB (1,000GB), each image has an average size of 145KB, one can use approximately 816GB for images. The remaining space accounts for around 184GB, which is usually allocated to system files, formatting, and other drive operations. The ample amount leaves room for many images to be stored while still allowing the system to run smoothly. That is an average size of 145 KB per image, meaning that this 816 GB allocation could store about 5,627,586 images.

As presented in Figure 46, the CCTV of the system was used to monitor and record events. It is a key component for the system to get data and record events. The camera enables real-time monitoring and compliance analysis.

[FIGURE 46: On-site deployment of the CCTV — photograph of CCTV camera mounted on a metal pole/post outdoors with overhead utility wires visible.]

Figure 46. On-site deployment of the CCTV

Figure 47 shows a PoE switch, which serves two primary functions. It provides power to the CCTV and acts as a bridge to establish a connection between the CCTV and the PC. This

connection is essential for the system to operate effectively, enabling data transmission and processing to generate results. This device is one of the core parts for the system to function. This PoE switch manages the connection of each device connected to it. When a Lan Cable is inserted into its ports, LED light will glow and blink if data is being sent or received on its port. This PoE switch has four (4) host ports and one uplink port wherein you can connect it to an internet, but configuration of the system further is highly suggested therefore basing things on its manual is highly recommended.

[FIGURE 47: PoE Switch of the System — photograph of small black PoE switch with green LED indicators and blue Ethernet cables connected.]

Figure 47. PoE Switch of the System

Figure 48 presents a Cat 6 UTP (Unshielded Twisted Pair) LAN cable, which serves as a connector wire or cable that enables the components of the system to connect and interact with each other. The LAN cable establishes a connection from the CCTV to the PoE switch and, finally, to the PC. It functions as the pathway for data transmission, similar to how a road facilitates the flow of vehicles. This cable ensures stable and high-speed data transfer, enabling seamless communication between the system's components and it ensures the reliability and efficiency of the network.

[FIGURE 48: Lan Cable of the System — close-up photograph of the LAN cable connection points highlighted with yellow box and red triangular callout markers.]

Figure 48. Lan Cable of the System

Software Development

The Helmet Recognition System (HRS) software architecture is modular and optimized for efficient data processing and analysis. The system's central part uses the YOLOv8 object detection algorithm, which is applied in a Python environment under Anaconda's control. This environment, named “yolov8_custom,” includes all libraries and dependencies required for the running of the detection software. The system is designed to process live video streams from IP-connected CCTV cameras via a PoE switch, and pre-recorded videos or static images. The

results of detection are also incorporated into a locally hosted website, fueled by XAMPP and a MySQL database, where admin can access, visualize, and manage data through a dynamic web dashboard. HTML and PHP scripts facilitate smooth communication between the backend and the user interface, providing real-time feedback on helmet detection accuracy and compliance.

As discussed in chapter 3 on page 62, the trained YOLOv8 object detection model detects motorcycles and helmets, classifying riders based on their helmet compliance. The system defines a Region of Interest (ROI), as shown by the yellow rectangular boundary in the image, where detections are analyzed and recorded. Additionally, the system can highlight areas with frequent non-compliance, helping authorities target enforcement efforts more effectively.

[FIGURE 49: CCTV footage with YOLOv8 System — annotated CCTV view with overlay green text “Motorcycle Count: 1, With Helmet: 1, Without Helmet: 0”, a motorcyclist with helmet inside a yellow ROI rectangular region, blue bounding box around the motorcycle, timestamp 11-18-2024 Mon 10:53:33 Camera 01.]

Figure 49. CCTV footage with YOLOv8 System

When a detected motorcycle crosses the ROI line, the system checks for a detected helmet's bounding box near the motorcycle's bounding box. If a helmet is detected with the predefined distance threshold from the motorcycle, it is classified as “with helmets.” Otherwise, it is classified as “without a helmet.”

The site features four menus: About, Dashboard, Live, and Directory. In Figure 50, the Dashboard menu displays all the tallied data, with each result represented in designated graphs. The data is organized by the day of the week, showing the total for each day and a total tally for the entire week. This provides an overview of the system's collected results in a clear, visual format.

[FIGURE 50: Dashboard Page of the System — screenshot of the system's web dashboard titled “Helmet Recognition System with Data Analytics” with three statistical panels: donut chart for Total Motorists (228), line chart for Total Violators (214), and bar chart for Total Non-Violators (14).]

Figure 50. Dashboard Page of the System

Figure 51 summarizes the system results in numerical form for 1992 motorists. Among those, 1833 motorists, or 92%, were classified as violators, and 159 motorists, or 8%, were categorized as not violating. This summary establishes the overall helmet compliance distribution across the monitored population, which is consistent with the proportions observed in the pie graph in Figure 52. The data highlights a high rate of violations, showing the need for better enforcement of helmet use. These results give useful information about the behavior of motorists and the effectiveness of helmet laws. Presenting the data in both numbers and a visual chart makes the compliance trends easier to understand.

[FIGURE 51: Total Motorists, Violators, and Non-Violators — three numeric panels showing Total Motorists: 1992, Total Violators: 1833, Total Non-Violators: 159.]

Figure 51. Total Motorists, Violators, and Non-Violators

[FIGURE 52: Total Distribution of the Motorists Diagram Result — donut chart titled "Distribution of Motorists" showing Non Violators (dark blue, ~8%) and Violators (light blue, ~92%).]

Figure 52. Total Distribution of the Motorists Diagram Result

Figure 52 shows in Pie Graph and presents graphically the relative number of violators and non-violators. The violators are shown as light blue, constituting 92% of the number, and the non-violators are dark blue, accounting for the remaining 8%. The relative number here directly corresponds to the number found in Figure 51, showing that the depiction is consistent.

Figure 53 plots the weekly trend in daily detection events for November 27 up to December 7. The graph is a line chart tracking the trend in motorcycles, violators, and non-violators from November 27 up to December 6, as mentioned earlier. This day is the peak weekly data shown in Figure 50. Significant violations are seen on Non-violators' counts were consistently low throughout the week, similar to the distribution between violators and non-violators, taken from Figures 51 and 52.

[FIGURE 53: Daily Distribution of the Motorists Diagram Result — line chart titled "Motorist Trends by Date" showing Motorcycle, Violators, and Non-Violators on dates 11-27 through 12-07, with Count on Y-axis (0-900). Motorcycle line peaks at ~800 on 12-06.]

Figure 53. Daily Distribution of the Motorists Diagram Result

Figure 54 shows the data on the motorist by week. There is a peak in total for the week of Dec 1-7 with violators well up at the top of each category represent in Bar Graph diagram. The weekly peak corresponds to the daily peak in Figure 52, around December 6. Figure 51 and Figure 52 are also aligned in that the data tends to be consistent in either different time intervals.

[FIGURE 54: Weekly Distribution of the Motorists Diagram Result — bar chart titled "Weekly Motorist Data" showing Motorists, Violators, Non-Violators counts (0-2,000) across four weeks: 11-24 to 11-30, 12-01 to 12-07, 12-08 to 12-14, 12-15 to 12-21. Peak in 12-01 to 12-07 week with Motorists ~1900 and Violators ~1800.]

Figure 54. Weekly Distribution of the Motorists Diagram Result

Figure 55 provides an overview of the Helmet Recognition System (HRS), highlighting its purpose as presented on the "About the System" page. This section explains the system's primary goal of ensuring helmet compliance among motorcycle riders by leveraging advanced technologies for real-time video analysis and detection.

While at In the About menu page, shown in Figure 56, users can find brief information about the system, including its purpose and functionality, as well as details about the researchers involved in the project. This section provides context and background for the system's development and implementation.

[FIGURE 55: Helmet Recognition System - About the System Page — screenshot of the web page showing "Helmet Recognition System for Road Safety" header with descriptive text about the system's purpose, image processing, YOLOv8, and traffic safety enforcement.]

Figure 55. Helmet Recognition System - About the System Page

[FIGURE 56: Helmet Recognition System - About Us Page — screenshot showing the "About Us" page with Technological University of the Philippines - Manila header, six profile photos with names of researchers: John Carl Castaneda, Joshua Mades, Robin Mosinde, Andrea Mae Quintana, Daniela Rivera, Paula Maria Sanidi.]

Figure 56. Helmet Recognition System - About Us Page

Figure 57 shows the Live menu tab, which allows the user to view the real-time footage captured by CCTV. This tab provides a live feed of what the CCTV is currently observing.

[FIGURE 57: Live Menu Tab of the System — screenshot showing "Real-Time YOLO Detection Stream" panel displaying CCTV live footage of motorcyclists on a road.]

Figure 57. Live Menu Tab of the System

Project Capabilities and Limitations

This project was built so local authorities could act against motorists who do not comply with the safety regulations for driving on the local road and highway. It enhances road safety by providing real-time monitoring and accurate data for better enforcement. Additionally, it supports local authorities in identifying trends and patterns in helmet compliance. The following are identified capabilities.

Capabilities

1. The recognition system can detect whether motorcycle riders wear a helmet or not when they enter the video frame of the CCTV Video Footage, and in the case of images, it can determine if the image of the rider has a helmet or not.

2. Realtime data analytics for the gathered results of the recognition system, which reports the daily and weekly detection by count and percentage through the local web host.

3. Recordings of the detected motorcycle by snapshot images and categorizing each of the gathered images.

4. Summarization of results gathered and projected with the use of visual aids or graphs, specifically the bar graph, line graph, and pie graph.

5. The portability of the system can be integrated into other devices, but performance varies according to the system unit's different specifications.

Limitations

1. Weather and lighting of the deployment location can hinder the accuracy and reliability of the system.

2. The quality of the video and monitoring footage affects the output of the recognition system.

3. The placement of the CCTV must be considered. The angle and placement of the camera make the system output less accurate, causing little to no detail of the pixel of the frame to be recognized, particularly in the head of the rider, and it is harder for the system to determine if one wears a helmet or not.

4. The system majorly focuses on motorcycles only. Inaccuracy of detection can still occur if YOLO thinks the object is the same as the one in the frame, even if it is not.

Project Testing Results

The testing phase of the helmet recognition system focused on evaluating its accuracy, efficiency, and real-time detection capabilities. The main objectives were to determine the system's ability to accurately detect helmet usage, assess its performance, and analyze its processing speed. The results are organized into key themes for clarity and conciseness, supported by statistical methods where appropriate.

Results of Training and Validation

Train/box_loss, train/cls_loss, and train/dfl_loss—all of them increased progressively over 30 epochs, which meant that the model was increasingly better at recognizing motorcycles and helmets. The declining losses in bounding boxes, classification, and focus loss for the model showed rising confidence in detecting helmets as the training continued. This shows that the model not only became better at accuracy as well as in predicting bounding boxes more accurately. The validation losses improved, ascertaining the model's capability to generalize very well to new data and its utility in real-world usage. Reference data can be found in Appendix F, page 130.

As demonstrated in Figure 58, train/box_loss reduced from 0.65776 to 0.454, indicating that bounding box prediction accuracy is better over time. The gentle, continuous change in loss curves, free from sudden drops or spikes, illustrates a solid training process capable of successfully reducing overfitting and instability risks (Aquitan et al., 2023).

[FIGURE 58: Training Losses for Bounding Box Regression — line chart titled “train/box_loss” with loss value 0.45-0.70 on Y-axis and epoch 0-20+ on X-axis. Shows declining trend from ~0.70 to ~0.45.]

Figure 58. Training Losses for Bounding Box Regression

This steady decline in loss is an indication that the model is learning well and gaining consistent improvements in its predictions. Furthermore, the fact that there are no sudden spikes

or dramatic dips in the loss curve indicates that the training is well-managed, resulting in more consistent and accurate results. This steady trend also signifies the model's capacity to generalize to unseen data, which is vital for real-world applications.

[FIGURE 59: Training Losses for Classification — line chart titled “train/cls_loss” with loss value 0.30-0.60 on Y-axis and epoch 0-20+ on X-axis. Shows declining trend from ~0.55-0.60 to ~0.30.]

Figure 59. Training Losses for Classification

The train/cls_loss measure in Figure 58 reveals a steep decline from 0.524 to 0.2921 (figure 59). The decline reflects the model's capacity to differentiate object classes more reliably. Early epochs see minor oscillations because the model starts learning, then there is a steady drop as it increases class discrimination. As the final epochs are reached, the loss levels stabilize, indicating that the model has learned enough to discriminate object classes.

[FIGURE 60: Training Losses for Distribution Focal Loss — line chart titled “train/dfl_loss” with loss value 0.90-1.05 on Y-axis. Declining trend from ~1.05 to ~0.90.]

Figure 60. Training Losses for Distribution Focal Loss

The train/dfl_loss, indicating the confidence of the model in making bounding box predictions, decreased from 1.04262 to 0.90809, indicating a consistent rise in prediction confidence (figure 60). Overall, these findings indicate that the YOLOv8 model has made significant improvements in helmet and motorcycle recognition during training since it also retains the steadiness and consistency of the trained model.

[FIGURE 61: Validation Losses for Bounding Box Regression — line chart titled “val/box_loss” with loss value 0.85-1.05 on Y-axis. Fluctuating declining trend from ~1.00 to ~0.85.]

Figure 61. Validation Losses for Bounding Box Regression

The val/box_loss showed a steady improvement over the course of the validation, with its value decreasing from 0.90258 to 0.84372 (figure 61). This decrease signifies that the model is becoming more capable of predicting bounding boxes, resulting in tighter and more precise object localization in unseen data.

The val/cls_loss in Figure 62 also had an increasing trend, falling from 0.52741 to 0.4889. This fall indicates the better performance of the model in object classification appropriately, exhibiting greater generalization when generalized to new data.

[FIGURE 62: Validation Losses for Classification Loss — line chart titled "val/cls_loss" with loss value 0.475-0.600 on Y-axis. Fluctuating declining trend from ~0.55-0.60 down to ~0.48.]

Figure 62. Validation Losses for Classification Loss

The val/dfl_loss in Figure 63 showed the greatest improvement, from 1.35383 to 0.90809. The decline signifies greater reliability in the distribution-based learning of the model, yielding better prediction and stable performance in validation settings.

[FIGURE 63: Validation Losses for Distribution Focal — line chart titled "val/dfl_loss" with loss value 1.30-1.50 on Y-axis. Fluctuating declining trend.]

Figure 63. Validation Losses for Distribution Focal

[FIGURE 64: Precision Metrics — line chart titled "metrics/precision(B)" with precision value 0.75-0.90 on Y-axis. Fluctuating trend.]

Figure 64. Precision Metrics

Based on the research of Aquitan et al. (2023), the performance measures of precision and recall, as false positives and false negatives, respectively, are helpful in assessing the adequacy of the YOLOv8 model (figure 63). As indicated by Precision Metrics in Figure 64, the proportion of correctly detected helmets and motorcycles to the total detected ones reached a maximum at epoch 17 with a value of 0.92842. This large value reflects that the model is effective in minimizing the rate of false detections.

[FIGURE 65: Recall Metrics — line chart titled "metrics/recall(B)" with recall value 0.84-0.94 on Y-axis.]

Figure 65. Recall Metrics

The metrics/recall(B) indicate the proportion of the number of actual helmets and motorcycles in the database that the YOLOv8 model was capable of finding (figure 65). This metric peaked at 0.95238 during the 12th epoch. The consistency in the model's high precision and recall metrics indicates its effectiveness in detecting objects accurately while false detections

and missed objects.

[FIGURE 66: Mean Average Precision at IoU threshold 0.5 Metrics — line chart titled "metrics/mAP50(B)" with mAP value 0.86-0.96 on Y-axis.]

Figure 66. Mean Average Precision at IoU threshold 0.5 Metrics

The mAP50 metric measures the YOLOv8 model's detection accuracy at a fixed Intersection over Union (IoU) threshold of 0.5 (figure 66). With a value of 0.94438, which stabilized around 0.90887, this high score reflects the model's strong ability to reliably detect, localize, and classify objects under moderate overlap criteria. These results highlight the model's overall detection reliability and effectiveness in scenarios where a less inflexible IoU threshold is sufficient for evaluation.

[FIGURE 67: Mean Average Precision at IoU threshold 0.5 to 0.95 Metrics — line chart titled "metrics/mAP50-95(B)" with mAP value 0.62-0.74 on Y-axis.]

Figure 67. Mean Average Precision at IoU threshold 0.5 to 0.95 Metrics

The mAP50-95 metric evaluates the model's performance across a range of IoU thresholds, from 0.5 to 0.95 at 0.05 increments. Initially valued at 0.65749, this metric (figure 67) showed progressive improvement, reaching 0.73442, featuring the model's enhanced precision and localization accuracy over time. This metric is stricter as it emphasizes performance at higher IoU thresholds, where accurate localization becomes increasingly challenging. The consistent improvement in the mAP50-95 metric shows the model's ability to perform well across varying levels of detection difficulty, confirming its robustness in handling complex scenarios.

During the validation phase, the confusion matrix shows a comprehensive overview of the YOLOv8 model's classification performance across the classes (figure 68). The confusion matrix was generated using the Confusion Matrix Generator developed by Dr. Damiano Perri, Ph.D.

[FIGURE 68: Confusion Matrix of the Model Validation Phase — 4x4 confusion matrix with TARGET (Motorcycle, Helmet, Head, Background) and OUTPUT axes. Diagonals: Motorcycle 14 (41.18%), Helmet 7 (20.59%), Head 6 (17.65%), Background 0 (0.00%). Precision: 87.50%, 63.64%, 100.00%, 0.00%. Recall: 100%, 100%, 85.71%, 100%. Overall: 27/34 = 79.41% Accuracy.]

Figure 68. Confusion Matrix of the Model Validation Phase

As shown in figure 68, the motorcycle class showed an actual positive value of 14 instances correctly classified as motorcycles (true positives), with two instances incorrectly classified as background. However, upon manually checking the validation results, the model correctly predicted motorcycles that were not included in the dataset labels for motorcycle class, classifying them as a part of the background class. Likewise, the helmet class correctly identified 7 helmets, but four were misclassified as background. This finding shows the capability of the model to correctly identify instances that were not included in the actual training dataset.

The head class had a perfect detection rate for the six instances of heads, with no misclassifications within this category. While the head class is part of the training, it is not the project's focus and is regarded as insignificant. Moreover, the background class presented some challenges, as one instance of the head was incorrectly predicted as a background, resulting in the actual head instance being undetected.

The confusion metrics emphasize the model's exceptional performance in important classes, particularly motorbike and helmet. This analysis emphasizes how improving training data and class differentiation is crucial to increasing the model's capacity to handle visually similar items in complex instances.

Table 9
Performance Metrics of the Trained YOLOv8 Model

Parameters	**Accuracy**	**Precision**	**Recall**	**Remarks**
Detected Motorcycle	79.4%	87.5%	100%	Passed
Detected Helmet	79.4%	63.6%	100%	Passed

The performance metrics presented above show the performance of the model, including the True Positives, False Positives, False Negatives, and True Negatives that have been acquired from the confusion matrix (figure 58). The calculations were based on the accuracy, precision, and recall formula (Perri, Simonetti, & Gervasi, 2022). While the overall dataset consists of 1,648 instances, the current training iteration (Train18), which will be used to evaluate the model,

utilizes a subset of 353 datasets trained over 30 epochs. The subset was selected to allow focused evaluation and optimization of the trained model.

Observations from the performance metrics demonstrated the capabilities of the YOLOv8 model in classification (table 9). With an accuracy of 79.4% for both the motorcycle and helmet classes, this fraction shows how the model is performing equally well in both classes. Nevertheless, the difference is evident with precision and recall. The precision values suggest some areas for improvement, with the motorcycle class attaining a high precision of 87.5%, and the helmet class falls behind with a precision of 63.6%. This difference shows that the model has difficulty distinguishing helmets from other examples, resulting in a greater number of false positives. Additionally, the 100% recall rate for both classes shows that the model is extremely good at identifying all examples of motorcycles and helmets in the dataset.

While the high precision and recall scores for both classes are evident, the lower precision of the YOLOv8 model for helmets illustrates the need for more optimization and more detailed dataset training. This implies more fine-tuning and a greater varied dataset would enhance the accuracy of the model for helmet detection.

Result of Testing Model

The YOLOv8 model was implemented and validated to match the key metrics for testing: accuracy, precision, and recall. An 8-hour deployment in a bright environment was captured on CCTV video for the testing. Figure 57 and Figure 58 show actual detection results produced by the model as seen through the camera.

[FIGURE 69: Live Stream Model Testing Result: Detection of Riders Wearing Helmets — annotated CCTV image of two riders on a motorcycle with green bounding boxes labeled “with Helmet ID: 12” and “with Helmet ID: 13” plus a blue bounding box labeled “Motorcycle ID: 5”.]

Figure 69. Live Stream Model Testing Result: Detection of Riders Wearing Helmets

As shown from Figure 69, the detection model identifies two riders wearing helmets. Each helmet is highlighted with labeled bounding boxes, this demonstrates the system's capability to detect and classify helmet compliance accurately.

Then in Figure 70, the detection model identifies a rider on a motorcycle labeled as “Motorcycle ID: 22.” The absence of helmet annotations indicates that the rider is not wearing a helmet, showcasing the model's effectiveness in detecting non-compliance with safety standards.

The researchers manually examined each class instance to determine true positive, false positive, true negative, and false negative values. Figure 71 displays an overview of the YOLOv8 model's classification performance across the classes during the model deployment phase using a Confusion Matrix Generator developed by Dr. Damiano Perri, Ph.D. This analysis allowed the researchers to assess the model's accuracy and identify areas for improvement. The confusion matrix provided valuable insights into misclassifications, highlighting patterns of errors and helping to refine the detection algorithm for better overall performance and reliability.

[FIGURE 70: Live Stream Model Testing Result: Detection of a Motorcyclist Without a Helmet — annotated image of motorcyclist on red motorcycle with blue bounding box labeled “Motorcycle ID: 22” - no helmet annotation.]

Figure 70. Live Stream Model Testing Result: Detection of a Motorcyclist Without a Helmet

[FIGURE 71: Confusion Matrix of Model Deployment — 3x3 confusion matrix with TARGET (Motorcycle, Helmet, Background) and OUTPUT axes. Motorcycle: 63 (47.01%), 4 (2.99%), 20 (14.93%), SUM 87 with 72.41% Precision. Helmet: 0 (0.00%), 26 (19.40%), 11 (8.21%), SUM 37 with 70.27% Precision. Background: 4 (2.99%), 6 (4.48%), 0 (0.00%), SUM 10 with 0.00% Precision. Recall: 94.03%, 72.22%, 100.00%. Overall: 89/134 = 66.42% Accuracy.]

Figure 71. Confusion Matrix of Model Deployment

Table 10

Performance Metrics of the Model Deployment

Parameters	Accuracy	Precision	Recall
Detected Motorcycle	66.42%	72.4%	94%
Detected Helmet	66.42%	70.27%	72.2%

Table 10 shows the performance metrics of 4 hours of CCTV footage using the data provided by the confusion matrix. The accuracy of both motorcycle and helmet achieved a ratio of 66.42%, indicating a consistent performance for both classes but hinting at further training.

The precision of both classes ranged from 70% to 72%, implying the model had a slightly harder time accurately recognizing helmets and motorcycles. The motorcycle class effectively recognized and classified most motorcycles in the footage with a strong recall of 94%. In contrast, the helmet class showed a recall of 72.2%. These results underline the YOLOv8 model's strengths in comprehensive recognition of classes and areas for improvement in precise classification.

Chapter 4

RESULTS AND DISCUSSION

The chapter presents the system design, which encompasses its structural elements, block diagrams, process flow diagrams, and step-by-step system development of the helmet detection system, merging the algorithm with data analytics. The operational procedures, testing methods, evaluation processes, system limitations, and future improvement recommendations are presented in this chapter.

Project Description

The helmet recognition system was developed to enhance road safety by ascertaining helmet compliance and if motorcycle riders are wearing helmets while riding a motorcycle. The system was deployed on public roads to minimize the risk of head injury and to monitor and ensure compliance.

The Helmet Recognition System was designed to enhance road safety by establishing helmet compliance and if motorcycle riders are wearing helmets while riding a motorcycle. The system was deployed on public roads to minimize the risk of head injuries and to track and enforce compliance. The system was intended to identify motorcycle helmet use by examining pre-recorded or live videos captured from roadways, intersections, highways, and particularly accident hotspots. Real-time or recorded video examination with HRS sought to enhance road safety by minimizing head injuries and crashes involving cyclists and pedestrians through effective monitoring and enforcement. The system consists of several important components: a monitor, a system unit, a CCTV camera, and a PoE switch. The CCTV camera serves to capture motorcycle riders in real-time. The video is forwarded to the system unit, where YOLOv8 processes and analyzes the video in real-time. According to YOLOv8's object detection capabilities, it recognizes whether the motorcyclist is wearing a helmet. The PoE switch provides power and data link to the CCTV camera, providing seamless and optimal operation without any

need for specific power sources. The data is transmitted to the system unit for analysis using the aid of Python programming language, which can support YOLOv8 in transforming detection outputs into data analytics. This module for analytics provides for more in-depth information, including trends in helmet compliance, such as helmet compliance patterns, reports, and other statistical information. The results are then displayed on the monitor for real-time monitoring and review.

[FIGURE 42: System Visual Output — two side-by-side images: (left) close-up photo of a CCTV camera mounted on a building exterior; (right) live system feed from the same camera showing a road scene with overlay green text reading "Motorcycle Count: 0, With Helmet: 0, Without Helmet: 0" and a yellow rectangular detection zone superimposed on the roadway.]

Figure 42. System Visual Output

Figure 42 shows the CCTV camera system integrated with a YOLO (You Only Look Once) object detection framework. The camera was installed on a nearby road. The video feed captured by the CCTV is processed by the YOLO-based system to detect and track objects, specifically motorcycles. The system analyzes the video in real-time, identifying whether the motorcycle riders are wearing helmets or not.

The system counted the number of motorcycles, and it detected if a rider is wearing a helmet or not. There is a marked yellow box where the system will focus on the detection area and process the objects it finds there.

Project Structure

The system comprises six major components: the PC, CCTV camera, monitor, PoE switch, and LAN cable. In terms of software, YOLOv8, Visual Studio, Anaconda Prompt, and XAMPP are implemented for the system's operation and functionality.

[FIGURE 43: PC used in the System — photograph of the inside of the desktop PC with two arrows labeling the major components: CPU (cooler/fans visible) and GPU (graphics card).]

Figure 43. PC used in the System

As shown in Figure 43, the PC serves as the brain of the system, hosting the YOLO program, storing gathered data, and acting as the server for the web interface displaying the

results. The CPU manages system operations, such as processing video frames, handling data flow, and running the web server. Meanwhile, the GPU accelerates the computationally intensive tasks of YOLO's neural network, enabling real-time object detection and classification. Together, the CPU and GPU ensure efficient processing of the CCTV feed, identifying motorcycles and detecting helmet usage accurately.

1. **Ryzen 5 3400g:** This AMD processor features 4 cores and 8 threads, along with integrated graphics. With a thermal design power (TDP) of only 65W, it is an energy-efficient choice, ideal for long-term system use.

2. **16GB DDR4 RAM:** This amount of memory is sufficient to run most applications without the risk of memory shortages, making it the optimal size for the program being used.

3. **GTX 1060 3G Variant:** Manufactured by NVIDIA, this GPU is suitable for general tasks and is fully capable of running the program without issues. It operates at 120W, making it an excellent choice for maintaining low energy costs.

4. **128GB SSD storage** used for Operating system which is Windows 10 and 1TB of HDD is reserved for the YOLOv8 dataset, training and all gathered data by the recognition system.

In Figure 44 shows the monitor, which is used to display information and events processed by the PC. It allows users to observe the system's progress and real-time activities, including the live feed from the CCTV and the results gathered from the data analysis.

[FIGURE 44: Live Footage of the CCTV via Monitor — photograph of a computer monitor displaying the system's interface with a live CCTV view of a road scene including a utility pole and trees.]

Figure 44. Live Footage of the CCTV via Monitor

Computation for the Resolution of the System (Reference for System Requirements)

Given:

1. 1 TB = 1,000,000,000 KB

2. Average size of image = 145 KB

number of images = total storage / average size of image

number of images = 1,000,000,000 / 145 ≈ 6,896,551 images

Figure 45. Number of Images per Storage Capacity Formula
Source: https://tinyurl.com/mrhhb3b6

According to Remington, G. (2023) in figure 45, the storage capacity for images depends on their average size. The calculation here assumes an image size of 145 KB. Image sizes vary based on resolution, format (JPEG, PNG, RAW), and compression. This estimate gives a rough idea of how many images can be stored on a 1 TB storage hard disk drive under these specific conditions.

Image Resolution

- Captured Image Resolution: 400×600

Video Resolution

- Input Video Resolution: 1920×1080 (Full HD)
- The System processes Full HD video at an average of 15 FPS, which is capable of handling high-quality streams with satisfactory performance. With the system unit having a total storage capacity of 1TB (1,000GB), each image has an average size of 145KB, one can use approximately 816GB for images. The remaining space accounts for around 184GB, which is usually allocated to system files, formatting, and other drive operations. The ample amount leaves room for many images to be stored while still allowing the system to run smoothly. That is an average size of 145 KB per image, meaning that this 816 GB allocation could store about 5,627,586 images.

As presented in Figure 46, the CCTV of the system was used to monitor and record events. It is a key component for the system to get data and record events. The camera enables real-time monitoring and compliance analysis.

[FIGURE 46: On-site deployment of the CCTV — photograph of a CCTV camera mounted on a metal pole/post outdoors with overhead utility wires visible against a partly cloudy sky.]

Figure 46. On-site deployment of the CCTV

Figure 47 shows a PoE switch, which serves two primary functions. It provides power to the CCTV and acts as a bridge to establish a connection between the CCTV and the PC. This

connection is essential for the system to operate effectively, enabling data transmission and processing to generate results. This device is one of the core parts for the system to function. This PoE switch manages the connection of each device connected to it. When a Lan Cable is inserted into its ports, LED light will glow and blink if data is being sent or received on its port. This PoE switch has four (4) host ports and one uplink port wherein you can connect it to an internet, but configuration of the system further is highly suggested therefore basing things on its manual is highly recommended.

[FIGURE 47: PoE Switch of the System — photograph of a small black PoE switch with green LED indicators lit and blue Ethernet cables connected.]

Figure 47. PoE Switch of the System

Figure 48 presents a Cat 6 UTP (Unshielded Twisted Pair) LAN cable, which serves as a connector wire or cable that enables the components of the system to connect and interact with each other. The LAN cable establishes a connection from the CCTV to the PoE switch and, finally, to the PC. It functions as the pathway for data transmission, similar to how a road facilitates the flow of vehicles. This cable ensures stable and high-speed data transfer, enabling seamless communication between the system's components and it ensures the reliability and efficiency of the network.

[FIGURE 48: Lan Cable of the System — close-up photograph of LAN cable connection points highlighted with a yellow box and red triangular callout markers showing the connections.]

Figure 48. Lan Cable of the System

Software Development

The Helmet Recognition System (HRS) software architecture is modular and optimized for efficient data processing and analysis. The system's central part uses the YOLOv8 object detection algorithm, which is applied in a Python environment under Anaconda's control. This environment, named “yolov8_custom,” includes all libraries and dependencies required for the running of the detection software. The system is designed to process live video streams from IP-connected CCTV cameras via a PoE switch, and pre-recorded videos or static images. The

results of detection are also incorporated into a locally hosted website, fueled by XAMPP and a MySQL database, where admin can access, visualize, and manage data through a dynamic web dashboard. HTML and PHP scripts facilitate smooth communication between the backend and the user interface, providing real-time feedback on helmet detection accuracy and compliance.

As discussed in chapter 3 on page 62, the trained YOLOv8 object detection model detects motorcycles and helmets, classifying riders based on their helmet compliance. The system defines a Region of Interest (ROI), as shown by the yellow rectangular boundary in the image, where detections are analyzed and recorded. Additionally, the system can highlight areas with frequent non-compliance, helping authorities target enforcement efforts more effectively.

[FIGURE 49: CCTV footage with YOLOv8 System — annotated CCTV view with overlay green text “Motorcycle Count: 1, With Helmet: 1, Without Helmet: 0”, a motorcyclist with helmet inside a yellow ROI rectangular region, blue bounding box around the motorcycle, with timestamp “11-18-2024 Mon 10:53:33 Camera 01”.]

Figure 49. CCTV footage with YOLOv8 System

When a detected motorcycle crosses the ROI line, the system checks for a detected helmet's bounding box near the motorcycle's bounding box. If a helmet is detected with the predefined distance threshold from the motorcycle, it is classified as “with helmets.” Otherwise, it is classified as “without a helmet.”

The site features four menus: About, Dashboard, Live, and Directory. In Figure 50, the Dashboard menu displays all the tallied data, with each result represented in designated graphs. The data is organized by the day of the week, showing the total for each day and a total tally for the entire week. This provides an overview of the system's collected results in a clear, visual format.

[FIGURE 50: Dashboard Page of the System — screenshot of the system's web dashboard titled “Helmet Recognition System with Data Analytics” with three statistical panels: a donut chart for Total Motorists (228), a line chart for Total Violators (214), and a bar chart for Total Non-Violators (14).]

Figure 50. Dashboard Page of the System

Figure 51 summarizes the system results in numerical form for 1992 motorists. Among those, 1833 motorists, or 92%, were classified as violators, and 159 motorists, or 8%, were categorized as not violating. This summary establishes the overall helmet compliance distribution across the monitored population, which is consistent with the proportions observed in the pie graph in Figure 52. The data highlights a high rate of violations, showing the need for better enforcement of helmet use. These results give useful information about the behavior of motorists and the effectiveness of helmet laws. Presenting the data in both numbers and a visual chart makes the compliance trends easier to understand.

[FIGURE 51: Total Motorists, Violators, and Non-Violators — three numeric panels showing Total Motorists: 1992, Total Violators: 1833, Total Non-Violators: 159.]

Figure 51. Total Motorists, Violators, and Non-Violators

[FIGURE 52: Total Distribution of the Motorists Diagram Result — donut chart titled "Distribution of Motorists" showing Non-Violators (dark blue, ~8%) and Violators (light blue, ~92%).]

Figure 52. Total Distribution of the Motorists Diagram Result

Figure 52 shows in Pie Graph and presents graphically the relative number of violators and non-violators. The violators are shown as light blue, constituting 92% of the number, and the non-violators are dark blue, accounting for the remaining 8%. The relative number here directly corresponds to the number found in Figure 51, showing that the depiction is consistent.

Figure 53 plots the weekly trend in daily detection events for November 27 up to December 7. The graph is a line chart tracking the trend in motorcycles, violators, and non-violators from November 27 up to December 6. This day is the peak weekly data shown in Figure 50. Significant violations are seen on Non-violators' counts were consistently low throughout the week, similar to the distribution between violators and non-violators, taken from Figures 51 and 52.

[FIGURE 53: Daily Distribution of the Motorists Diagram Result — line chart titled "Motorist Trends by Date" showing Motorcycle, Violators, and Non-Violators on dates 11-27 through 12-07, with Count on Y-axis (0-900). The Motorcycle line peaks at ~800 on 12-06.]

Figure 53. Daily Distribution of the Motorists Diagram Result

Figure 54 shows the data on the motorist by week. There is a peak in total for the week of Dec 1–7 with violators well up at the top of each category represent in Bar Graph diagram. The weekly peak corresponds to the daily peak in Figure 52, around December 6. Figure 51 and Figure 52 are also aligned in that the data tends to be consistent in either different time intervals.

[FIGURE 54: Weekly Distribution of the Motorists Diagram Result — bar chart titled "Weekly Motorist Data" showing Motorists, Violators, Non-Violators counts (0-2,000) across four weeks: 11-24 to 11-30, 12-01 to 12-07, 12-08 to 12-14, 12-15 to 12-21. Peak in 12-01 to 12-07 week with Motorists ~1,900 and Violators ~1,800.]

Figure 54. Weekly Distribution of the Motorists Diagram Result

Figure 55 provides an overview of the Helmet Recognition System (HRS), highlighting its purpose as presented on the "About the System" page. This section explains the system's primary goal of ensuring helmet compliance among motorcycle riders by leveraging advanced technologies for real-time video analysis and detection.

While at In the About menu page, shown in Figure 56, users can find brief information about the system, including its purpose and functionality, as well as details about the researchers involved in the project. This section provides context and background for the system's development and implementation.

[FIGURE 55: Helmet Recognition System - About the System Page — screenshot of the web page showing the "Helmet Recognition System for Road Safety" header with descriptive text about the system's purpose, image processing, YOLOv8, and traffic safety enforcement.]

Figure 55. Helmet Recognition System - About the System Page

[FIGURE 56: Helmet Recognition System - About Us Page — screenshot of the "About Us" page with Technological University of the Philippines - Manila header, College of Industrial Technology, Bachelor of Engineering Technology Major in Computer Engineering Technology, and six profile photos with names of the researchers: John Carl Castaneda, Joshua Mades, Robin Mosinde, Andrea Mae Quintana, Daniela Rivera, Paula Maria Sanidi.]

Figure 56. Helmet Recognition System - About Us Page

Figure 57 shows the Live menu tab, which allows the user to view the real-time footage captured by CCTV. This tab provides a live feed of what the CCTV is currently observing.

[FIGURE 57: Live Menu Tab of the System — screenshot showing "Real-Time YOLO Detection Stream" panel displaying CCTV live footage of motorcyclists on a road.]

Figure 57. Live Menu Tab of the System

Project Capabilities and Limitations

This project was built so local authorities could act against motorists who do not comply with the safety regulations for driving on the local road and highway. It enhances road safety by providing real-time monitoring and accurate data for better enforcement. Additionally, it supports local authorities in identifying trends and patterns in helmet compliance. The following are identified capabilities.

Capabilities

1. The recognition system can detect whether motorcycle riders wear a helmet or not when they enter the video frame of the CCTV Video Footage, and in the case of images, it can determine if the image of the rider has a helmet or not.

2. Realtime data analytics for the gathered results of the recognition system, which reports the daily and weekly detection by count and percentage through the local web host.

3. Recordings of the detected motorcycle by snapshot images and categorizing each of the gathered images.

4. Summarization of results gathered and projected with the use of visual aids or graphs, specifically the bar graph, line graph, and pie graph.

5. The portability of the system can be integrated into other devices, but performance varies according to the system unit's different specifications.

Limitations

1. Weather and lighting of the deployment location can hinder the accuracy and reliability of the system.

2. The quality of the video and monitoring footage affects the output of the recognition system.

3. The placement of the CCTV must be considered. The angle and placement of the camera make the system output less accurate, causing little to no detail of the pixel of the frame to be recognized, particularly in the head of the rider, and it is harder for the system

to determine if one wears a helmet or not.

4. The system majorly focuses on motorcycles only. Inaccuracy of detection can still occur if YOLO thinks the object is the same as the one in the frame, even if it is not.

Project Testing Results

The testing phase of the helmet recognition system focused on evaluating its accuracy, efficiency, and real-time detection capabilities. The main objectives were to determine the system's ability to accurately detect helmet usage, assess its performance, and analyze its processing speed. The results are organized into key themes for clarity and conciseness, supported by statistical methods where appropriate.

Results of Training and Validation

Train/box_loss, train/cls_loss, and train/dfl_loss—all of them increased progressively over 30 epochs, which meant that the model was increasingly better at recognizing motorcycles and helmets. The declining losses in bounding boxes, classification, and focus loss for the model showed rising confidence in detecting helmets as the training continued. This shows that the model not only became better at accuracy as well as in predicting bounding boxes more accurately. The validation losses improved, ascertaining the model's capability to generalize very well to new data and its utility in real-world usage. Reference data can be found in Appendix F, page 130.

As demonstrated in Figure 58, train/box_loss reduced from 0.65776 to 0.454, indicating that bounding box prediction accuracy is better over time. The gentle, continuous change in loss curves, free from sudden drops or spikes, illustrates a solid training process capable of successfully reducing overfitting and instability risks (Aquitan et al., 2023).

[FIGURE 58: Training Losses for Bounding Box Regression — line chart titled “train/box_loss” with loss value 0.45-0.70 on Y-axis and epoch 0-20+ on X-axis. Shows declining trend from ~0.70 to ~0.45.]

Figure 58. Training Losses for Bounding Box Regression

This steady decline in loss is an indication that the model is learning well and gaining consistent improvements in its predictions. Furthermore, the fact that there are no sudden spikes or dramatic dips in the loss curve indicates that the training is well-managed, resulting in more consistent and accurate results. This steady trend also signifies the model's capacity to generalize to unseen data, which is vital for real-world applications.

[FIGURE 59: Training Losses for Classification — line chart titled “train/cls_loss” with loss value 0.30-0.60 on Y-axis and epoch 0-20+ on X-axis. Shows declining trend from ~0.55-0.60 to ~0.30.]

Figure 59. Training Losses for Classification

The train/cls_loss measure in Figure 58 reveals a steep decline from 0.524 to 0.2921 (figure 59). The decline reflects the model's capacity to differentiate object classes more reliably. Early epochs see minor oscillations because the model starts learning, then there is a steady drop as it increases class discrimination. As the final epochs are reached, the loss levels stabilize, indicating that the model has learned enough to discriminate object classes.

[FIGURE 60: Training Losses for Distribution Focal Loss — line chart titled “train/dfl_loss” with loss value 0.90-1.05 on Y-axis. Shows declining trend from ~1.05 to ~0.90.]

Figure 60. Training Losses for Distribution Focal Loss

The train/dfl_loss, indicating the confidence of the model in making bounding box predictions, decreased from 1.04262 to 0.90809, indicating a consistent rise in prediction confidence (figure 60). Overall, these findings indicate that the YOLOv8 model has made significant improvements in helmet and motorcycle recognition during training since it also retains the steadiness and consistency of the trained model.

[FIGURE 61: Validation Losses for Bounding Box Regression — line chart titled “val/box_loss” with loss value 0.85-1.05 on Y-axis. Shows fluctuating declining trend from ~1.00 to ~0.85.]

Figure 61. Validation Losses for Bounding Box Regression

The val/box_loss showed a steady improvement over the course of the validation, with its value decreasing from 0.90258 to 0.84372 (figure 61). This decrease signifies that the model is becoming more capable of predicting bounding boxes, resulting in tighter and more precise

object localization in unseen data.

The val/cls_loss in Figure 62 also had an increasing trend, falling from 0.52741 to 0.4889. This fall indicates the better performance of the model in object classification appropriately, exhibiting greater generalization when generalized to new data.

[FIGURE 62: Validation Losses for Classification Loss — line chart titled "val/cls_loss" with loss value 0.475-0.600 on Y-axis. Shows fluctuating declining trend from ~0.55-0.60 down to ~0.48.]

Figure 62. Validation Losses for Classification Loss

The val/dfl_loss in Figure 63 showed the greatest improvement, from 1.35383 to 0.90809. The decline signifies greater reliability in the distribution-based learning of the model, yielding better prediction and stable performance in validation settings.

[FIGURE 63: Validation Losses for Distribution Focal — line chart titled "val/dfl_loss" with loss value 1.30-1.50 on Y-axis. Shows fluctuating declining trend.]

Figure 63. Validation Losses for Distribution Focal

[FIGURE 64: Precision Metrics — line chart titled "metrics/precision(B)" with precision value 0.75-0.90 on Y-axis. Shows fluctuating trend with peak at ~0.92.]

Figure 64. Precision Metrics

Based on the research of Aquitan et al. (2023), the performance measures of precision and recall, as false positives and false negatives, respectively, are helpful in assessing the adequacy of the YOLOv8 model (figure 63). As indicated by Precision Metrics in Figure 64, the proportion of correctly detected helmets and motorcycles to the total detected ones reached a maximum at epoch 17 with a value of 0.92842. This large value reflects that the model is effective in minimizing the rate of false detections.

[FIGURE 65: Recall Metrics — line chart titled "metrics/recall(B)" with recall value 0.84-0.94 on Y-axis. Shows fluctuating trend with peak at ~0.95.]

Figure 65. Recall Metrics

The metrics/recall(B) indicate the proportion of the number of actual helmets and motorcycles in the database that the YOLOv8 model was capable of finding (figure 65). This

metric peaked at 0.95238 during the 12th epoch. The consistency in the model's high precision and recall metrics indicates its effectiveness in detecting objects accurately while false detections and missed objects.

[FIGURE 66: Mean Average Precision at IoU threshold 0.5 Metrics — line chart titled "metrics/mAP50(B)" with mAP value 0.86-0.96 on Y-axis.]

Figure 66. Mean Average Precision at IoU threshold 0.5 Metrics

The mAP50 metric measures the YOLOv8 model's detection accuracy at a fixed Intersection over Union (IoU) threshold of 0.5 (figure 66). With a value of 0.94438, which stabilized around 0.90887, this high score reflects the model's strong ability to reliably detect, localize, and classify objects under moderate overlap criteria. These results highlight the model's overall detection reliability and effectiveness in scenarios where a less inflexible IoU threshold is sufficient for evaluation.

[FIGURE 67: Mean Average Precision at IoU threshold 0.5 to 0.95 Metrics — line chart titled "metrics/mAP50-95(B)" with mAP value 0.62-0.74 on Y-axis. Shows progressive improvement over epochs.]

Figure 67. Mean Average Precision at IoU threshold 0.5 to 0.95 Metrics

The mAP50-95 metric evaluates the model's performance across a range of IoU thresholds, from 0.5 to 0.95 at 0.05 increments. Initially valued at 0.65749, this metric (figure 67) showed progressive improvement, reaching 0.73442, featuring the model's enhanced precision and localization accuracy over time. This metric is stricter as it emphasizes performance at higher IoU thresholds, where accurate localization becomes increasingly challenging. The consistent improvement in the mAP50-95 metric shows the model's ability to perform well across varying levels of detection difficulty, confirming its robustness in handling complex scenarios.

During the validation phase, the confusion matrix shows a comprehensive overview of the YOLOv8 model's classification performance across the classes (figure 68). The confusion matrix was generated using the Confusion Matrix Generator developed by Dr. Damiano Perri, Ph.D.

[FIGURE 68: Confusion Matrix of the Model Validation Phase — 4×4 confusion matrix with TARGET (Motorcycle, Helmet, Head, Background) on horizontal axis and OUTPUT on vertical axis. Diagonal: Motorcycle 14 (41.18%), Helmet 7 (20.59%), Head 6 (17.65%), Background 0

(0.00%). Right column shows precision: 87.50%, 63.64%, 100.00%, 0.00%. Bottom row shows recall: 100%, 100%, 85.71%, 100%. Overall: 27/34 = 79.41% Accuracy.]

Figure 68. Confusion Matrix of the Model Validation Phase

As shown in figure 68, the motorcycle class showed an actual positive value of 14 instances correctly classified as motorcycles (true positives), with two instances incorrectly classified as background. However, upon manually checking the validation results, the model correctly predicted motorcycles that were not included in the dataset labels for motorcycle class, classifying them as a part of the background class. Likewise, the helmet class correctly identified 7 helmets, but four were misclassified as background. This finding shows the capability of the model to correctly identify instances that were not included in the actual training dataset.

The head class had a perfect detection rate for the six instances of heads, with no misclassifications within this category. While the head class is part of the training, it is not the project's focus and is regarded as insignificant. Moreover, the background class presented some challenges, as one instance of the head was incorrectly predicted as a background, resulting in the actual head instance being undetected.

The confusion metrics emphasize the model's exceptional performance in important classes, particularly motorbike and helmet. This analysis emphasizes how improving training data and class differentiation is crucial to increasing the model's capacity to handle visually similar items in complex instances.

Table 9
Performance Metrics of the Trained YOLOv8 Model

Parameters	**Accuracy**	**Precision**	**Recall**	**Remarks**
Detected Motorcycle	79.4%	87.5%	100%	Passed
Detected Helmet	79.4%	63.6%	100%	Passed

The performance metrics presented above show the performance of the model, including the True Positives, False Positives, False Negatives, and True Negatives that have been acquired from the confusion matrix (figure 58). The calculations were based on the accuracy, precision,

and recall formula (Perri, Simonetti, & Gervasi, 2022). While the overall dataset consists of 1,648 instances, the current training iteration (Train18), which will be used to evaluate the model, utilizes a subset of 353 datasets trained over 30 epochs. The subset was selected to allow focused evaluation and optimization of the trained model.

Observations from the performance metrics demonstrated the capabilities of the YOLOv8 model in classification (table 9). With an accuracy of 79.4% for both the motorcycle and helmet classes, this fraction shows how the model is performing equally well in both classes. Nevertheless, the difference is evident with precision and recall. The precision values suggest some areas for improvement, with the motorcycle class attaining a high precision of 87.5%, and the helmet class falls behind with a precision of 63.6%. This difference shows that the model has difficulty distinguishing helmets from other examples, resulting in a greater number of false positives. Additionally, the 100% recall rate for both classes shows that the model is extremely good at identifying all examples of motorcycles and helmets in the dataset.

While the high precision and recall scores for both classes are evident, the lower precision of the YOLOv8 model for helmets illustrates the need for more optimization and more detailed dataset training. This implies more fine-tuning and a greater varied dataset would enhance the accuracy of the model for helmet detection.

Result of Testing Model

The YOLOv8 model was implemented and validated to match the key metrics for testing: accuracy, precision, and recall. An 8-hour deployment in a bright environment was captured on CCTV video for the testing. Figure 57 and Figure 58 show actual detection results produced by the model as seen through the camera.

[FIGURE 69: Live Stream Model Testing Result: Detection of Riders Wearing Helmets — annotated CCTV image showing two riders on a motorcycle, with green bounding boxes labeled “with Helmet ID: 12” and “with Helmet ID: 13”, and a blue bounding box labeled “Motorcycle ID: 5”.]

Figure 69. Live Stream Model Testing Result: Detection of Riders Wearing Helmets

As shown from Figure 69, the detection model identifies two riders wearing helmets. Each helmet is highlighted with labeled bounding boxes, this demonstrates the system's capability to detect and classify helmet compliance accurately.

Then in Figure 70, the detection model identifies a rider on a motorcycle labeled as “Motorcycle ID: 22.” The absence of helmet annotations indicates that the rider is not wearing a helmet, showcasing the model's effectiveness in detecting non-compliance with safety standards.

The researchers manually examined each class instance to determine true positive, false positive, true negative, and false negative values. Figure 71 displays an overview of the YOLOv8 model's classification performance across the classes during the model deployment phase using a Confusion Matrix Generator developed by Dr. Damiano Perri, Ph.D. This analysis allowed the researchers to assess the model's accuracy and identify areas for improvement. The confusion matrix provided valuable insights into misclassifications, highlighting patterns of errors and helping to refine the detection algorithm for better overall performance and reliability.

[FIGURE 70: Live Stream Model Testing Result: Detection of a Motorcyclist Without a Helmet — annotated image of a motorcyclist on a red motorcycle with blue bounding box labeled “Motorcycle ID: 22”. No helmet annotation is shown, indicating non-compliance.]

Figure 70. Live Stream Model Testing Result: Detection of a Motorcyclist Without a Helmet

[FIGURE 71: Confusion Matrix of Model Deployment — 4×4 confusion matrix with TARGET (Motorcycle, Helmet, Background) on horizontal axis and OUTPUT on vertical axis. Motorcycle row: 63 (47.01%), 4 (2.99%), 20 (14.93%), SUM 87 with 72.41% Precision. Helmet row: 0 (0.00%), 26 (19.40%), 11 (8.21%), SUM 37 with 70.27% Precision. Background row: 4 (2.99%), 6 (4.48%), 0 (0.00%), SUM 10 with 0.00% Precision. Bottom row recall: 94.03%, 72.22%, 100.00%. Overall: 89/134 = 66.42% Accuracy.]

Figure 71. Confusion Matrix of Model Deployment

Table 10

Performance Metrics of the Model Deployment

Parameters	Accuracy	Precision	Recall
Detected Motorcycle	66.42%	72.4%	94%
Detected Helmet	66.42%	70.27%	72.2%

Table 10 shows the performance metrics of 4 hours of CCTV footage using the data provided by the confusion matrix. The accuracy of both motorcycle and helmet achieved a ratio of 66.42%, indicating a consistent performance for both classes but hinting at further training. The precision of both classes ranged from 70% to 72%, implying the model had a slightly harder time accurately recognizing helmets and motorcycles. The motorcycle class effectively recognized and classified most motorcycles in the footage with a strong recall of 94%. In contrast, the helmet class showed a recall of 72.2%. These results underline the YOLOv8 model's strengths in comprehensive recognition of classes and areas for improvement in precise classification.

Project Evaluation Results

It passed through evaluation to ensure the helmet identification system could be used, operate, and work well in real-world applications. Major objectives such as system dependability, user experience, and overall efficiency are the basis of the review process. Results are themed for clarity, with tables summarizing average ratings and descriptions.

After conducting the testing procedures, the researchers surveyed thirty (30) respondents, including motorcycle riders, people who know traffic management, local government officials, and professionals with technical expertise. The prototype was assessed using a survey form that follows the ISO 25010:2011 criteria. These are Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, and Portability. The participants responded based on a rating scale provided by the Likert scale; therefore, responses ranged from 1 being the lowest to 5 being the highest. The mean scores for each question were calculated to provide a more in-depth evaluation of the system's performance based on respondents' feedback, and the following results were very good/very acceptable across all the criteria.

Table 11
Summary of Evaluation Results of the Functional Suitability Criteria

Criterion	Sub-Criteria	Rating	Interpretation

Functional Suitability	The set of functions covers all the specified tasks and user objectives.	4.43	Very Good
	The system provides the correct results with the needed degree of precision.	4.20	Very Good
	The system facilitates the accomplishment of specified tasks and objectives.	4.36	Very Good
	Overall	**4.33**	**Very Good**

Table 11 provides a comprehensive summary of the evaluation results for the functional suitability criteria. The system's performance across three key sub-criteria was assessed: the completeness sub-criterion, which was rated 4.43, the correctness sub-criterion, which received a rating of 4.20, and the appropriateness sub-criterion, which scored 4.36. All three sub-criteria fall within the “Very Good” scale, demonstrating the system's strong functional suitability. The overall average rating of 4.33 also places the system within the “Very Good” scale.

Table 12
Summary of Evaluation Results of the Performance Efficiency Criteria

Criterion	Sub-Criteria	Rating	Interpretation
Performance Efficiency	The response time throughout rates of the system when performing its functions meet requirements.	4.66	Excellent
	Degree to which the amounts and types of resources used by a product or system, when performing its functions, meet requirements.	4.50	Very Good
	Degree to which the maximum limits of the products or system, parameter meet requirements.	4.73	Excellent
	Overall	**4.63**	**Excellent**

Table 12 presents the summary of the evaluation results for the performance efficiency criteria. The time behavior sub-criterion received a rating of 4.66, indicating strong performance in responsiveness. The resource utilization sub-criterion was rated 4.50, reflecting efficient use of

system resources. The capacity sub-criterion scored 4.73, showing excellent scalability. The overall average rating of 4.63 places the system within the “Excellent” scale, highlighting its strong performance efficiency.

Table 13
Summary of Evaluation Results of the Compatibility Criteria

Criterion	Sub-Criteria	Rating	Interpretation
Compatibility	Degree to which a product can perform its required functions efficiently while sharing common environment and resources with other products, without detrimental impact on any other product.	4.80	Excellent
	Degree to which two or more systems, products or components can exchange information and use the information that has been exchanged.	4.87	Excellent
	Accuracy and completeness with which users achieve specified goals.	4.73	Excellent
	Overall	**4.80**	**Excellent**

The evaluation results for the compatibility criteria are summarized in Table 13. The system's performance was assessed across three key sub-criteria: co-existence, which received a rating of 4.80, interoperability, which was rated 4.87, and effectiveness, which was scored 4.73. All three sub-criteria fall within the “Excellent” scale, reflecting the system's outstanding performance in compatibility. The overall average rating of 4.80 places the system firmly within the “Excellent” scale. These results indicate that the system seamlessly integrates with existing technologies and functions efficiently without conflicts. The high ratings confirm the system's reliability with other platforms.

Table 14
Summary of Evaluation Results of the Usability Criteria

Criterion	Sub-Criteria	Rating	Interpretation

Usability	The system has attributes that make it easy to operate and control.	4.80	Excellent
	The system can be used by specified users of the product.	4.83	Excellent
	The system can be used by people with the widest range of characteristics and capabilities to achieve a specified goal in a specified context of use.	4.37	Very Good
	Overall	**4.67**	**Excellent**

Table 14 presents the summary of the usability evaluation results for the Helmet Recognition System with Algorithm Analysis. The system received a rating of 4.80 for ease of use, placing it in the “Excellent” scale, 4.83 for user suitability, also on the “Excellent” scale, and 4.37 for accessibility, which falls in the “Very Good” scale. The overall average rating of 4.67 places the system in the “Excellent” scale, indicating that it effectively meets usability standards.

Table 15
Summary of Evaluation Results of the Reliability Criteria

Criterion	Sub-Criteria	Rating	Interpretation
Reliability	The system is operational when required for use.	4.70	Excellent
	The system is accessible when required for use.	4.76	Excellent
	The system meets needs for reliability under normal operation.	4.80	Excellent
	Overall	**4.75**	**Excellent**

Table 15 presents the summary of the reliability evaluation results. The system's performance was assessed across three key sub-criteria: operational availability, accessibility, and reliability under normal operation. The system received a rating of 4.70 for operational availability, 4.76 for accessibility, and 4.80 for reliability under normal operation. All three sub-criteria fall within the “Excellent” scale, demonstrating the system's outstanding reliability. The overall average rating of 4.75, which also falls within the “Excellent” scale, highlights the

system's consistent and dependable performance, ensuring its reliability for the intended use and its ability to operate effectively over time.

Table 16
Summary of Evaluation Results of the Portability Criteria

Criterion	Sub-Criteria	Rating	Interpretation
Portability	The system can effectively and efficiently be adapted for different or evolving hardware, software, or other operational or usage environments.	4.57	Excellent
	The system can be successfully installed and/or uninstalled in a specified environment.	4.47	Very Good
	Degree to which a product can replace another specified software product for the same purpose in the same environment.	4.70	Excellent
	Overall	**4.58**	**Excellent**

Table 16 shows the evaluation results for portability. The system received a rating of 4.57 for adaptability to different hardware, 4.47 for its installation and removal across various environments, and 4.70 for its ability to replace another software product for the same purpose. The overall average rating is 4.58, which falls within the “Excellent” scale.

Table 17
Summary of Evaluation Results

Criteria	Mean	Interpretation
Functional Suitability	4.33	Very Good
Performance Efficiency	4.63	Excellent
Compatibility	4.80	Excellent
Usability	4.67	Excellent
Reliability	4.75	Excellent
Portability	4.58	Excellent

Average Evaluation Grade	**4.63**	**Excellent**

Table 17 summarizes the evaluation results across all assessed criteria, including Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, and Portability, alongside their respective average ratings. After computing the overall evaluation, the system achieved an average appraisal mark of 4.63, which falls within the “Excellent” scale. These results indicate that the system consistently meets high standards across multiple quality attributes.

Table 18
Summary of Cronbach's Alpha Result

Description	Values	Internal Consistency
No. of Items	18	
Sum of Variance	4.76	
Variance Total Score	15.44	
Cronbach's α	**0.73**	

Then, Cronbach's Alpha was computed, which resulted in a score of 0.73 with the sample population in survey which is 30 respondents, as presented in Table 18. It is in the range that signifies “Acceptable” internal consistency.

Table 19
Validity and Reliability of Construct

Factor	Min	Max	Mean	Cronbach's Alpha
Perceived of Usefulness	3	5	4.32	0.688
Perceived ease of Use	3	5	4.68	0.574
System Adaptable	3	5	4.63	0.631
The behavior of Intention to Use	4	5	4.8	0.625

Table 19 presents the reliability analysis of various factors using Cronbach's alpha. The results indicate that the reliability scores range from 0.574 to 0.688. The values fall within the

questionable and poor categories. Specifically, “Perceived Usefulness,” “System Adaptability,” and “The Behavior of Intention to Use” exhibit questionable reliability, while “Perceived Ease of Use” falls into the poor category.

[FIGURE 72: Result of Internal Consistency of Helmet Recognition System With Data Analysis — horizontal bar chart showing Cronbach's Alpha Score from 0.000 to 1.000 with bars for: Average Cronbach's Alpha 0.730, Perceived of Usefulness 0.688, Perceived ease of Use 0.574, System Adaptable 0.631, The behavior of Intention to Use 0.625.]

Figure 72. Graphical Representation of Cronbach

These findings suggest that while some factors demonstrate moderate consistency, further refinement or reassessment may be necessary to enhance reliability. The survey result is found on Appendix E, page 131.

Chapter 5

SUMMARY OF FINDINGS, CONCLUSIONS, AND RECOMMENDATIONS

This chapter presents the results of the design, development, testing, and evaluation procedures conducted to realize the objectives of the research. Specifically, the discussion points include the summary of findings, conclusions, and recommendations.

Summary of Findings

The primary goal of this research was to develop a helmet detection system using image processing and machine learning algorithms, specifically YOLOv8, to enhance road safety by monitoring and ensuring compliance with helmet-wearing regulations. The system captures real-time footage using CCTV, captured images, and recorded videos, and it processes it to identify helmet usage among motorcycle riders.

The following are the key findings of this study:

1. The “Helmet Recognition System with Algorithm Analysis” was designed and implemented using various hardware components, including a CCTV camera, PC, monitor, PoE switch, and LAN cables, integrated with YOLOv8 for real-time image processing.

2. The hardware setup, including the Ryzen 5 3400g processor, GTX 1060 3G GPU, and 16GB RAM, provided sufficient computational power to run the system without issues, offering real-time analysis and a low-cost solution in terms of energy consumption.

3. The system's testing demonstrated reliable detection accuracy for both helmeted and non-helmeted riders, but the CCTV used by the researchers does not output the same reliability as the testing environment.

4. As per the internal consistency evaluation results, respondents rated the system with a Cronbach's Alpha value of 0.72, indicating “Acceptable” reliability. The survey consisted of 18 items using a 5-point Likert scale, with the sum of the item variance at 5.22 and the variance of the total score at 16.2. Overall, the system is perceived as “Acceptable”, demonstrating consistent responses across all items evaluated.

Conclusions

Derived from the outcomes of the project investigation in relation to the study objectives, the researchers reached the following conclusions:

1. The helmet detection system successfully meets its primary objective of identifying motorcycle riders without helmets and analyzing their compliance with road safety rules.

2. The designed image recognition system, leveraging the YOLOv8 object detection algorithm, accurately distinguishes helmeted from non-helmeted riders under various conditions, although performance may vary in challenging environments like poor lighting, adverse weather, and footage quality. It also detects some types of vehicles that are two-wheeled and look like one, such as tricycles.

3. The implemented data analytics tools proved effective in processing and quantifying daily and weekly data on helmet usage, providing valuable insights for local authorities to improve road safety measures.

4. Evaluations conducted using the ISO 25010:2011 model revealed strong performance in terms of functional suitability, performance efficiency, and usability but identified potential improvements in areas of reliability under extreme environmental conditions.

Recommendations

As an improvement to the identified limitations of the project study, the researchers proposed the following recommendations:

1. Improve algorithm performance in low-light and harsh weather conditions.

2. Expand datasets for better detection of various helmet types.

3. Expand datasets for other types of two-wheeled vehicles such as bikes, tricycles, e-bikes, and more to enable the system to identify and not detect the same category.

4. Account for additional factors such as passengers or multiple riders.

5. Explore the integration of cloud-based processing for scalability.

6. Expand to other safety gear detection and broader road safety monitoring.

7. Use a capable CCTV that can output very high-resolution footage.

8. Enforce the strategic deployment of CCTV cameras in key areas, such as highways, intersections, and accident-prone zones.

REFERENCES

[Editor's note: The References section (originally pages 119–121 of the source thesis) was not included in the photograph upload and is therefore not reproduced here. Citations referenced throughout the body text include: Aquitan et al. (2023), Giron et al. (2020, 2022), Susa et al. (2022), Swapna et al. (2019), Talaulikar et al. (2019), Dasgupta et al. (2019), Abbas et al. (2012), Bachani et al. (2017), Dela Cruz et al. (2022), Deng et al. (2020), Nantulya and Reich (2003), Sy (2017), Tan (2018), Shuey (2019), Padmini et al. (2020), Xu et al. (2018), Chang et al. (2020), Chai et al. (2021), Tian et al. (2024), Takahashi (2016), Alshamrani and Battab (2015), Perri, Simonetti, & Gervasi (2022), Remington (2023), and the World Health Organization (2023). The full APA-style reference list should be inserted here from the original document.]

APPENDICES

Appendix A

GANTT CHART

[GANTT CHART (large multi-row schedule with three sub-tables): (1) Calendar of Activities for the Initial Phase covering Group Formation (Feb 27, 2024), Brainstorming for Possible Topic (Feb 28 – Mar 5, 2024), Three Chapter 1 Topics for Topic Proposal (Mar 1 – Mar 17, 2024), Topic Proposal (Apr 18, 2024). (2) Chapter 1 (Introduction) sub-tasks: Background of the Study (May 5–6), Objective of the Study (May 7–8), Scope and Limitations of the Study (May 9–10), Significance of the Study (May 11–12), Chapter 1 Final Draft (May 22–27). (3) Chapter 2 (Conceptual Framework): Topic Outline (May 7–8), Review of Related Literature and Studies (May 8–22), Conceptual Model (May 22–24), Operational Definition of Terms (May 22–24), Chapter 2 Final Draft (May 26 – Jun 1). (4) Chapter 3 (Research Methodology): Project Design (May 25–27), Project Development (May 27–29), Operation and Testing Procedure (May 29 – Jun 1), Chapter 3 Final Draft (Jun 2–7). (5) Later phases through September–November 2024 (week-by-week Gantt bars): Chapters 1–3 Revision, Chapter 4 Drafting, Chapter 5 Drafting, Project Design, Project Development, Project Testing, Project Deployment, Project Evaluation, Chapter 4–5 Revision, Chapter 1–5 Finalization.]

Appendix B

PROPOSED BUDGETARY REQUIREMENTS

Hardware Requirements

Materials	Price	Quantity	Total
CCTV Camera	Php 2,200.00	1	Php 2,200.00
LAN Cable	Php 900.00	1 (60 Meters)	Php 900.00
PoE Switch	Php 1,200.00	1	Php 1,200.00
Monitor	Php 1,500.00	1	Php 1,500.00
System Unit			Php 8,000.00
Ryzen 5 3400g	Php 2,000.00	1	
a320 Motherboard	Php 1,000.00	1	
16gb DDR4 RAM	Php 1,000.00	1	
GTX 1060 3G	Php 2,000.00	1	
450W Power Supply	Php 500.00	1	
128G SSD	Php 500.00	1	
1TB HDD	Php 1,000.00	1	
Case	Php 0.00	1	
		Net Total:	**Php 13,800.00**

Software Requirements

Materials	Price	Quantity	Total
Windows 10 (Unlicensed)	Free	1	Php 0.00
YOLOv8	Free	1	Php 0.00
Anaconda Prompt	Free	1	Php 0.00
Visual Studio	Free	1	Php 0.00
PyCharm	Free	1	Php 0.00
CVAT.ai	Free	1	Php 0.00
		Net Total:	**Php 0.00**

Appendix C

PICTURES TAKEN DURING FABRICATION, TESTING AND EVALUATION

A. FABRICATION

Researchers then proceed to create and develop the system after researching the necessary procedures by using related literature, tutorials and more to be able to accomplish its objectives.

[PHOTO: Researchers seated together with laptops working on the system.]

The researchers use tutorials and later will be used for development.

[PHOTO: Researchers handling Hikvision CCTV equipment with cables on a coding/testing setup.]

The researchers code and test using the proposed equipment and materials.

B. TESTING

After the researcher's successful fabrication, they now start deploying the system to the deployment site to conduct and see results which later be evaluated.

[PHOTO: Researchers wiring the CCTV camera at the deployment site outdoors near a wall.]

The researchers are wiring the CCTV camera which will be used to gather live footage for the detection system.

[PHOTO: Researchers installing the CCTV on a metal fence/post on a rooftop with neighborhood view.]

The researchers hooked up the CCTV camera and lined up the preferred camera angle that can output a reliable data source.

[PHOTO: Monitor showing the CCTV footage of the deployment scene.]

The researchers then check the footage of the CCTV in the monitor to ensure that it is working properly and start the detection process.

C. EVALUATION

The researchers present the following features and functionalities of the project prototype to the respondents and authorities of the deployment location.

[PHOTO: A researcher demonstrating the system to a respondent on a tablet.]

[PHOTO: Researchers with local citizens and authorities outdoors conducting a survey.]

The researchers presented and surveyed the local citizens and authorities of the project. Sir Eric S. Tierra and Sir Philip Lasheras assist the researchers in conducting a survey within its municipality.

[PHOTO: Group picture of researchers with MDRRMO staff inside their office.]

Group Picture of researchers and the Municipal Disaster Risk Reduction Management Office (MDRRMO) at Atimonan, Quezon. Researchers are very much appreciated the assistance and to the following persons:

- Eric S. Tierra (IT Specialist in MDRRMO)
- Philip Lasheras (Technical Specialist in MDRRMO)
- Ma. Christie Balitaon (Brgy. Captain of Atimonan, Quezon at Brgy. Zone 1)
- Luigi Martinez (CCTV Operator in MDRRMO)

And also, to the people who participated in our survey that given us proprietary data needed in the research.

Appendix D

SAMPLE OF EVALUATION INSTRUMENT

Evaluation Instrument for Helmet Recognition System with Algorithm Analysis

Name (Optional): _____ Date of Evaluation: _____

Profession: _____

Instructions: Kindly encircle the appropriate scale that corresponds to the evaluation of the project using the scale below:

Numerical Scale	Equivalent Rating
5	Excellent
4	Very Good
3	Good
2	Fair
1	Poor

Indicators (with Likert Scale 5–1 columns):

Functional Suitability

Completeness. The set of functions covers all the specified tasks and user objectives.

Correctness. The system provides the correct results with the needed degree of precision.

Appropriateness. The system facilitates the accomplishment of specified tasks and objectives.

Performance Efficiency

Time behavior. The response time throughput rates of the system when performing its functions meet requirements.

Resource Utilization. Degree to which the amounts and types of resources used by a product or system, when performing its functions, meet requirements.

Capacity. Degree to which the maximum limits of the products or system, parameter meet requirements.

Compatibility

Co-existence. Degree to which a product can perform its required functions efficiently while sharing common environment and resources with other products, without detrimental impact on any other product.

Interoperability. Degree to which two or more systems, products or components can exchange information and use the information that has been exchanged.

Effectiveness. Accuracy and completeness with which users achieve specified goals.

Usability

Operability. The system has attributes that make it easy to operate and control.

Learnability. The system can be used by specified users of the product.

Accessibility. The system can be used by people with the widest range of characteristics and capabilities to achieve a specified goal in a specified context of use.

Reliability

Availability (operational). The system is operational when required for use.

Availability (accessible). The system is accessible when required for use.

Maturity. The system meets needs for reliability under normal operation.

Portability

Adaptability. The system can effectively and efficiently be adapted for different or evolving hardware, software or other operational or usage environments.

Install-ability. The system can be successfully installed and/or uninstalled in a specified environment.

Replaceability. Degree to which a product can replace another specified software product for the same purpose in the same environment.

Further comments regarding the system:

_____

Appendix E

RESULT OF THE ASSESSMENT FOR THE CRONBACH'S ALPHA

Table for Cronbach's Alpha with Computation

	K (# of questions)	E-var	Variance Total Score	Cronbach's Alpha
SU (Functional Suitability)	3	0.99	1.82	0.69
SEU (Performance Efficiency / Usability)	6	1.49	2.86	0.57
SA (Compatibility)	6	1.90	4.01	0.63
IU (Reliability / Portability)	3	0.37	0.64	0.63

Min / Max / Mean per Category

Category	Min	Max	Mean
Functional Suitability	3	5	4.32
Performance Efficiency	3	5	4.63
Usability	3	5	4.68
Compatibility	4	5	4.80
Portability	3	5	4.58
Reliability	3	5	4.76

Summary of the Survey

Counts of Strongly Agree (5) / Agree (4) / Neutral (3) / Disagree (2) / Strongly Disagree (1) across all 18 indicators, given by 30 respondents:

Survey Question	SA(5)	A(4)	N(3)	D(2)	SD(1)
The set of functions covers all the specified tasks and user objectives.	13	17	0	0	0
The system provides the correct results with the needed degree of precision.	9	18	3	0	0
The system facilitates the accomplishment of specified tasks and objectives.	12	17	1	0	0
Response time throughput rates of the system meet requirements.	21	9	0	0	0
Resource utilization meets requirements.	17	11	2	0	0

Maximum limits of the products or system parameter meet requirements.	22	8	0	0	0
Product can perform required functions efficiently sharing environment with others.	24	6	0	0	0
Two or more systems can exchange information and use the information exchanged.	26	4	0	0	0
Accuracy and completeness with which users achieve specified goals.	22	8	0	0	0
The system has attributes that make it easy to operate and control.	24	6	0	0	0
The system can be used by specified users of the product.	25	5	0	0	0
The system can be used by people with widest range of characteristics and capabilities.	14	13	3	0	0
The system is operational when required for use.	21	9	0	0	0
The system is accessible when required for use.	23	7	0	0	0
The system meets needs for reliability under normal operation.	24	6	0	0	0
The system can be adapted for different or evolving hardware/software environments.	19	9	2	0	0
The system can be successfully installed/uninstalled in a specified environment.	18	8	4	0	0
A product can replace another specified software product for the same purpose.	22	7	1	0	0

Appendix F

RAW DATA FOR TRAINING AND VALIDATION PERFORMANCE METRICS

Training Metrics by Epoch (1–30)

epoch	time	train/box_loss	train/cls_loss	train/dfl_loss	metrics/precision(B)	metrics/recall(B)	metrics/mAP50(B)	metrics/mAP50-95(B)
1	2722.5	0.65776	0.524	1.0426	0.84673	0.85714	0.94438	0.6574
2	4845.7	0.68918	0.5612	1.0588	0.84096	0.95183	0.96447	0.6742
3	7412.6	0.69782	0.5538	1.0607	0.79549	0.94601	0.96597	0.6976
4	10107	0.72928	0.5883	1.0718	0.75231	0.89854	0.87112	0.6219
5	12694	0.72904	0.5756	1.0634	0.75369	0.89755	0.87922	0.6649
6	14916	0.71935	0.5733	1.0713	0.84751	0.94509	0.95445	0.6595
7	16935.3	0.70394	0.5726	1.06678	0.8242	0.90476	0.94092	0.6756
8	19119.2	0.69239	0.56048	1.05988	0.79734	0.9008	0.86811	0.64629
9	21149.1	0.69452	0.55017	1.06315	0.79746	0.88671	0.89244	0.67604
10	23109.6	0.68413	0.54159	1.05249	0.79153	0.89919	0.90609	0.70129
11	25607.2	0.67396	0.52442	1.0496	0.91288	0.89553	0.967	0.68432
12	27835.2	0.66007	0.52371	1.0409	0.83105	0.95238	0.95946	0.71329
13	29877	0.66111	0.51859	1.04254	0.8105	0.85714	0.88387	0.67022
14	32078.8	0.63874	0.49796	1.03164	0.80465	0.85714	0.86289	0.69104
15	34224.7	0.65023	0.50121	1.03051	0.81659	0.85714	0.89181	0.68233
16	36218.3	0.65215	0.50099	1.04302	0.82685	0.83312	0.88919	0.69686
17	38320.2	0.62707	0.48372	1.01884	0.92842	0.90476	0.97063	0.7035
18	40442.4	0.62055	0.46315	1.01769	0.80931	0.90476	0.88088	0.6933
19	42679.9	0.61484	0.46965	1.017	0.79428	0.90226	0.86858	0.691
20	44940.2	0.5942	0.44095	1.00803	0.79505	0.901	0.88018	0.698
21	46901.2	0.5383	0.37432	0.94131	0.78567	0.90064	0.87225	0.703
22	49089.2	0.52042	0.3578	0.94469	0.8103	0.8915	0.89312	0.71
23	51354.9	0.51185	0.34565	0.93426	0.79446	0.90147	0.88458	0.7
24	53713.4	0.51454	0.36724	0.93862	0.8642	0.85714	0.89332	0.70265
25	55714.1	0.49177	0.32331	0.92475	0.8626	0.83068	0.89943	0.71312
26	58283.8	0.4806	0.31413	0.9152	0.86296	0.85714	0.88535	0.70255
27	60204	0.46162	0.29968	0.90393	0.83046	0.82701	0.90886	0.71339
28	62375.9	0.46576	0.29887	0.90879	0.85714	0.85714	0.8959	0.72938
29	64343.5	0.45911	0.298	0.90596	0.82842	0.85714	0.90852	0.73668
30	66242.9	0.454	0.2921	0.90809	0.83048	0.85714	0.90887	0.73442

Validation Metrics by Epoch (1–30)

epoch	time	val/box_loss	val/cls_loss	val/dfl_loss	lr/pg0	lr/pg1	lr/pg2
1	2722.5	0.90258	0.52741	1.35383	0.000473	0.000473	0.000473
2	4845.75	0.9606	0.53387	1.33582	0.000918	0.000918	0.000918
3	7412.66	0.95379	0.54861	1.36375	0.001332	0.001332	0.001332

4	10107.4	1.04102	0.58218	1.50288	0.001288	0.001288	0.001288
5	12694.9	1.03797	0.60321	1.43196	0.00124	0.00124	0.00124
6	14916.6	1.05289	0.60253	1.43656	0.001193	0.001193	0.001193
7	16935.3	0.99246	0.54754	1.36279	0.001146	0.001146	0.001146
8	19119.2	1.03745	0.56032	1.38153	0.001099	0.001099	0.001099
9	21149.1	0.98507	0.50558	1.33972	0.001052	0.001052	0.001052
10	23109.6	0.92094	0.50822	1.28812	0.001005	0.001005	0.001005
11	25607.2	0.94808	0.51776	1.39841	0.000957	0.000957	0.000957
12	27835.2	0.88122	0.52815	1.34267	0.00091	0.00091	0.00091
13	29877	0.92386	0.52626	1.32815	0.000863	0.000863	0.000863
14	32078.8	0.92072	0.52011	1.29891	0.000816	0.000816	0.000816
15	34224.7	0.94255	0.52151	1.33667	0.000769	0.000769	0.000769
16	36218.3	0.92302	0.49814	1.3526	0.000722	0.000722	0.000722
17	38320.2	0.88428	0.48295	1.36403	0.000674	0.000674	0.000674
18	40442.4	0.88594	0.50468	1.34739	0.000627	0.000627	0.000627
19	42679.9	0.93709	0.51752	1.41158	0.00058	0.00058	0.00058
20	44940.2	0.87565	0.50148	1.36182	0.000533	0.000533	0.000533
21	46901.2	0.83464	0.46334	1.33644	0.000486	0.000486	0.000486
22	49089.2	0.84638	0.48859	1.39588	0.000439	0.000439	0.000439
23	51354.9	0.86857	0.47771	1.38745	0.000392	0.000392	0.000392
24	53713.4	0.88553	0.49965	1.41304	0.000344	0.000344	0.000344
25	55714.1	0.85842	0.48807	1.38316	0.000297	0.000297	0.000297
26	58283.8	0.85426	0.4894	1.35952	0.00025	0.00025	0.00025
27	60204	0.87111	0.52948	1.40139	0.000203	0.000203	0.000203
28	62375.9	0.86283	0.49271	1.42816	0.000156	0.000156	0.000156
29	64343.5	0.8621	0.4869	1.40704	0.000109	0.000109	0.000109
30	66242.9	0.84372	0.4889	1.3693	6.14e-05	6.14e-05	6.14e-05

Appendix G

PROFILE OF RESPONDENTS

No.	Name	Company / Institution	Position / Designation
1	Eric S. Tierra	Municipal Disaster Risk Reduction and Management	IT Specialist
2	Philip Lasheras	Municipal Disaster Risk Reduction and Management	Technical Specialist
3	Luigi Martinez	Municipal Disaster Risk Reduction and Management	CCTV Operator
4	Ma. Christie Balitaon	Barangay Zone 1 Poblacion, Atimonan, Quezon	Barangay Captain
5	Benjamin Baute Jr.	Traffic Management Unit	Traffic Management Administrator
6	N/A	Adamson University	Student
7	N/A	Adamson University	Student
8	Elijah San Luis	SEAGATE	Engineering Specialist
9	Raniel De Vera	Technological University of the Philippines - Manila	Student
10	Frank Thomas Trinidad	University of Caloocan City	Student
11	Eneriano Monje	University of Caloocan City	Student
12	N/A	Universidad de Manila	Student
13	N/A	Universidad de Manila	Student
14	N/A	Pamantasan ng Lungsod ng Maynila	Student
15	N/A	Universidad de Manila	Student
16	Elson Deleon	Foodpanda	Delivery Rider
17	Eduardo Latinazo	Foodpanda	Delivery Rider
18	Alvin Charles	Polytechnic University of the Philippines	Student
19	Bernadette Abarquez	Polytechnic University of the Philippines	Student
20	Ajay Dacanay	Joyride	MC Taxi Driver
21	Roderick Balang	Smartly	IT Support
22	Mark Galang	Joyride	MC Taxi Driver
23	Joshua Sabado	Joyride	MC Taxi Driver
24	Monsi Sulapas	JC Consultants, Inc.	Lead Security
25	Julian Teodoro	Rephil Station	CCTV Operator

26	Eduardo Miranda	Rephil Station	Surveillance Technician
27	Patrick Dianito	Teleperformance	Technician
28	Earl Jacque Barretto	Universidad de Manila	Student
29	Zyrell Avila	Universidad de Manila	Student
30	Joanna Bartolome	Universidad de Manila	Student

Appendix H

RESEARCHERS' DEPLOYMENT LETTER

[DEPLOYMENT LETTER (on Technological University of the Philippines letterhead, dated October 30, 2024) addressed to Ms. Christie Ricafrente Balitaon, Barangay Captain, Barangay Zone I Poblacion, Atimonan, Quezon 4331.]

October 30, 2024

Ms. Christie Ricafrente Balitaon

Barangay Captain

Barangay Zone I Poblacion

Atimonan, Quezon 4331

Dear Ma'am/Sir,

We are a group of 4th-year students of Bachelor of Engineering Technology major in Computer Engineering Technology at the Technological University of the Philippines - Manila. As part of our academic requirements, we are undertaking an undergraduate thesis project titled “Helmet Recognition System with Algorithm Analysis.” We are therefore writing to seek permission to implement this project in Barangay Zone I Poblacion, Atimonan, Quezon.

The aim of our project is to promote safety on the roads by monitoring the helmet on users who ride motorcycles around the barangay. In this regard, we plan to install a CCTV system on a specific position and will collect the data over time for two (2) to three (3) days, with tentative dates set for November 9–11. This process is essential for assessing helmet compliance within the barangay, providing insights that support road safety ideas and evaluating the effectiveness of our system.

We assure you that we will adhere strictly to all relevant protocols and respect the privacy of residents throughout the data collection process. Furthermore, we are committed to conducting this study with the utmost professionalism and diligence.

We kindly request your support and guidance in facilitating the deployment of our system. We would be grateful for the opportunity to meet with you at your convenience to

discuss the project in detail and address any questions or concerns you may have.

Thank you for considering our request. We look forward to your positive response.

Sincerely yours,

Prepared by:

JOHN CARL C. CASTAÑEDA (TUPM-21-1601) JOSHUA C. MADES (TUPM-21-3092)

ROBIN C. MOSINDE (TUPM-21-3093) ANDREA MAE S. QUINTANA (TUPM-21-5074)

DANIELA G. RIVERA (TUPM-21-3380) PAULA MARIE G. SANIEL (TUPM-21-3336)

Noted by:

MARC ARDIE ARDIENTE — Project Study Adviser

DR. RALPH SHERWIN CORPUZ — Project Development Professor

Appendix I

CERTIFICATION OF GRAMMARIAN

[OFFICIAL CERTIFICATION DOCUMENT (formatted certificate page with signature).]

THESIS GRAMMARIAN CERTIFICATION

This is to certify that the master thesis entitled,

HELMET RECOGNITION SYSTEM WITH ALGORITHM ANALYSIS

authored by

JOHN CARL C. CASTAÑEDA

JOSHUA C. MADES

ROBIN C. MOSINDE

ANDREA MAE S. QUINTANA

DANIELA G. RIVERA

PAULA MARIE G. SANIEL

has undergone editing and proofreading by the undersigned.

This Certification is being issued upon their request for whatever purposes it may serve them.

JENNIFER P. ALINSUNOD, CHRA, LPT, MA-English

Grammarian

Technological University of the Philippines

January 20, 2025

Date of Issuance

Appendix J

CERTIFICATION OF TURNITIN

[OFFICIAL CERTIFICATE OF SIMILARITY INDEX USING TURNITIN, on Technological University of the Philippines letterhead (VRE-URD).]

CERTIFICATE OF SIMILARITY INDEX USING TURNITIN

This is to certify that the manuscript entitled

"**HELMET RECOGNITION SYSTEM WITH ALGORITHM ANALYSIS**"

authored by

JOHN CARL C. CASTAÑEDA

JOSHUA C. MADES

ROBIN C. MOSINDE

ANDREA MAE S. QUINTANA

DANIELA G. RIVERA

PAULA MARIE G. SANIEL

has been subjected to a similarity check on **February 17, 2025** using Turnitin with generated similarity index of **13%**.

Processed by:

DENNIS J. TABUCOL

Faculty Staff, URDS

Certified correct by:

Engr. HERONAFINE C. DE GUZMAN

OIC-Director, URDS

Document Details: Submission ID: trn:oid:25992:82388130. Submission Date: Feb 17, 2025, 1:45 PM GMT+8. File: Castañeda-HELMET-RECOGNITION-SYSTEM-WITH-ALGORITHM-ANALYSIS-with-revision.docx, 18.5 MB. 166 Pages, 22,622 Words, 127,108 Characters.

Match Groups: 250 Not Cited or Quoted (13%); 0 Missing Quotations; 0 Missing Citation; 0 Cited and Quoted. **Top Sources:** 9% Internet sources; 5% Publications; 11% Submitted works (Student Papers). **Integrity Flags:** 0 flags for review — no suspicious text manipulations found.

Appendix K

RECOGNITION SYSTEM'S MANUAL

HELMET RECOGNITION SYSTEM WITH ALGORITHM ANALYSIS — MANUAL

Table of Contents:

Section	Title	Page
I.	Things you'll need	1
II.	Connecting the PoE Switch to the PC	1
III.	Connecting the IP CCTV Camera to the PoE Switch	2
IV.	Setting up the Camera on the PC (Reference Manual from Hikvision)	2
V.	Binding the IP CCTV to the Recognition System	4
VI.	Startup of the YOLOv8 Recognition System	5
VII.	Accessing data gathered of the Recognition System	7
VIII.	Maintenance	10

I. Things you'll need:

- IP CCTV camera(s) (Max number can be used depends on the number of PoE LAN ports availability)
- PoE switch
- PC
- Ethernet cables (at least Cat5e or higher)
- System Partition with D: drive directory

[MANUAL FIGURE 1: Helmet Recognition System Network Layout — diagram showing PC connected via green cable to a PoE switch, with three CCTV cameras connected via yellow cables.]

Manual Figure 1. Helmet Recognition System Network Layout

[MANUAL FIGURE 2: Hikvision DS-3E0505P-E (Unmanaged PoE Switch) — photo of blue 5-port PoE switch.]

Manual Figure 2. Hikvision DS-3E0505P-E (Unmanaged PoE Switch)

II. Connecting the PoE Switch to the PC:

- Plug an LAN cable into one of the network ports on the PoE switch. As can be seen in Manual Figure 1, insert a LAN cable preferably the port no. 1 on the PoE port category.
- Connect the other end of the Ethernet cable to the Ethernet port on your PC.

III. Connecting the IP CCTV Camera to the PoE Switch:

- Place your IP CCTV Camera to your preferred location but for this recognition to work properly and accurately make sure to place the CCTV just high enough where motorcycles and its driver is visible and better with a good lighting condition.
- Connect another LAN cable from the IP CCTV camera to an available port on the PoE switch. Refer to the figure below which is indicated where to put the LAN cable on the CCTV.

[MANUAL FIGURE 3: Hikvision HiWatch HWI-B140H 4MP — photo of white CCTV camera.]

Manual Figure 3. Hikvision HiWatch HWI-B140H 4MP

- The PoE switch will supply power and network connectivity to the camera through the LAN cable just make sure the cable is cat 5e or above.

IV. Setting up the Camera on the PC (Reference Manual from Hikvision):

IP CCTV always comes with activation before user can be able to access the device and always make sure to refer to the manufacturer's manual of the CCTV. In this case we will use the Hikvision user manual as the system uses its CCTV.

Activate Device via Web Browser

Use web browser to activate the device.

Before You Start: Make sure your device and your PC connect to the same LAN.

- Open a web browser and input the default IP address. The default IP address of the device is 192.168.1.64.
- Create and confirm the admin password. It is recommended to make a STRONG PASSWORD (using a minimum of 8 characters, including upper case letters, lower

case letters, numbers, and special characters) and make sure not to forget it.

[MANUAL FIGURE 4: Activation — screenshot of Hikvision activation dialog.]

Manual Figure 4. Activation

- Click OK to complete activation and enter Live View page.
- Modify IP address of the camera:

1. Enter IP address modification page. Configuration → Network → TCP/IP

2. Change IP address. It is better to make a Static IP address and not use DHCP for convenience for later.

3. Save the settings.

[MANUAL FIGURE 5: Hikvision CCTV Configuration GUI — screenshot of TCP/IP configuration UI.]

Manual Figure 5. Hikvision CCTV Configuration GUI

- CCTV Device is now ready for use for the Recognition System. In case of more than one (1) IP CCTV, just repeat the activation process. For more information and further exploration of the IP CCTV functionality, please refer to the manufacturer's manual: https://www.hikvision.com/ph/products/IP-Products/Network-Cameras/Value-Series/DS-2CD1043G0-I/

V. Binding the IP CCTV to the Recognition System:

- Install the following application via internet or use of other device and transfer to the PC that is connected to the CCTV:

1. PyCharm (link: https://www.jetbrains.com/pycharm/). *Note: you may also use other software that can read and write python program file.*

In case of wanting to have internet in the PC, connect the modem or router to the port no. 5 or the uplink port based on Manual Figure 2 and make sure the IP of the PC and the CCTV is under the network address of the modem or router. Example: If modem or router's IP address is 192.168.1.1 subnet mask of 255.255.255.0, PC and CCTV address must be under 192.168.1.N where N is the interchangeable value except 1 since it is reserve to router and under the same subnet mask.

- Open the file of “main.py” within the recognition directory using the software you just installed.

[MANUAL FIGURE 6: CCTV configuration line — code snippet showing main.py with cctv_url variable and area definition for ROI. Highlights: red text = must not change; green = username; yellow = password (with colon between green and yellow); light blue = IP address (with @ symbol between yellow and light blue).]

Manual Figure 6. CCTV configuration line

- Under line 39 as can be seen in Manual Figure 6, change it to the respective CCTV profile.
- All the red highlighted text is must not change.
- The green highlighted text is the username you made when you first tried to activate the CCTV.
- The yellow highlighted text is the password. Don't forget to add colon (":") between the green and yellow highlighted text.
- Lastly is the light blue highlighted text is the IP of the CCTV you have settled. Also don't forget to include "@" symbol in between the yellow and light blue highlighted text.
- To add another CCTV just add another line and do the same thing as line 39 and make sure it is under the line 39 too.

VI. Startup of the YOLOv8 Recognition System

- Install Anaconda Prompt to the PC (Link: https://www.anaconda.com/download). The software will ask you to provide an email address to be able to download their software but you can always skip that part under the submit button.
- After installing, open the software and input the following commands:

D: — this command tells the prompt to go to the D drive directory. In case the recognition system is under the OS directory simply change it into "C:". If system is under C drive, use "cd \" instead to navigate backward since it defaulted to the Users Directory. We recommend to place the software under D drive for convenience.

CD Detect — this command directs the directory of the system to the file location of the recognition system. *Detect* is the name of the folder or directory used by researchers; in case it is different, change the character value to the folder's name instead.

conda activate yolov8_custom — this command is necessary to startup the system. The screen will be changed into the video footage gathered by the CCTV in real time.

[MANUAL FIGURE 7: Anaconda Prompt — screenshot of Anaconda Prompt window in initial state.]

Manual Figure 7. Anaconda Prompt

[MANUAL FIGURE 8: Helmet Recognition System Sample Footage — screenshot showing CCTV footage with overlay detection labels (Motorcycle Count, With Helmet, Without Helmet, timestamp 11-18-2024 Mon 10:54:55 Camera 01) and yellow ROI polygon.]

Manual Figure 8. Helmet Recognition System Sample Footage

VII. Accessing data gathered of the Recognition System

- Install the following software to be able to see the results of the gathered information by the system: **XAMPP**. This software is necessary to run the website locally.

- Open the software and start Apache and MySQL. Refer to Manual Figure 9.

[MANUAL FIGURE 9: XAMPP Control Panel — screenshot of XAMPP v3.3.0 with Apache and MySQL services running.]

Manual Figure 9. XAMPP Control Panel

- Place all the .php files under the XAMPP directory. To know how to place the folder file of the website, navigate to **C:\xampp\htdocs** if the software is installed as every setting on default and simply paste the folder wherein all the web files are inside. Refer to Manual Figure 10.

- Open any browser application installed on the PC and simply paste the following: **C:/xampp/htdocs/dashboard.php**. Now you are able to see the results of the data gathered by the system and all results.

[MANUAL FIGURE 10: Website Directory — screenshot of file explorer showing the website folder in C:\xampp\htdocs.]

Manual Figure 10. Website Directory

[MANUAL FIGURE 11: About Page — screenshot of the system's About page showing six researcher profile photos.]

Manual Figure 11. About Page

[MANUAL FIGURE 12: Dashboard Page — screenshot of the system's dashboard showing donut chart, line chart, and bar chart with motorist data.]

Manual Figure 12. Dashboard Page

[MANUAL FIGURE 13: Live Feed Page — screenshot of the live CCTV camera feed page.]

Manual Figure 13. Live Feed Page

VIII. Maintenance

- To maintain the system, we only highly recommend to just simply delete all the gathered images of the Recognition system. Simply go to this directory:

C:/xampp/htdocs/Helmet — where all subfolders can be found

C:/xampp/htdocs/Helmet/non_violators — folder for the non-violator images

C:/xampp/htdocs/Helmet/violators — folder for violators images

C:/xampp/htdocs/Helmet/motorcycles — folder for motorcycle images

and delete everything or just delete the images that is past 15–30 days. We also suggest using a higher capacity storage drive to be able to store for a longer period of time.

[MANUAL FIGURE 14: Image Captured Directory — screenshot of file explorer showing many small thumbnail images organized in directories.]

Manual Figure 14. Image Captured Directory